Minimal Complex Projective Surfaces

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a senior thesis submitted in partial fulfillment
of the requirements for the degree of
Bachelor of Arts in Mathematics at
Princeton University

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April 29, 2026

Abstract

We give an exposition of the classification of minimal models of smooth complex projective surfaces, with an emphasis on the interplay between complex analytic and complex algebraic methods. We begin with an introduction to the intersection form for complex algebraic surfaces, and show that the topological intersection form on four-manifolds agrees with the algebraic intersection form on complex projective surfaces. We then discuss the structure of birational maps of smooth surfaces, where we show every birational map can be built out particularly simple birational maps called blowups. This allows us to make precise the notion of a minimal model of a surface S, a "simplest" representative of the birational equivalence class of S, and we work to characterize the structure of these minimal models and determine when a surface has a unique minimal model.

In the case of irrational ruled surfaces, we show that every minimal model is a geometrically ruled surface, and in the case of rational surfaces, we show that all minimal models are either $\mathbb{P}^2$ or a Hirzebruch surface. When S is not ruled or rational, we show that there is a unique minimal model in it's birational equivalence class. Finally, we interpret these classification results through the lens of the Minimal Model Program, and give a brief discussion on the birational classification problem in higher dimensions.

Acknowledgements

First and foremost, I want to give my deepest thanks to my advisor, Prof. János Kollár, for all of his guidance throughout this thesis. This project would not have been possible without his support. I have learned so much from the process of working through this thesis, and his patience, feedback, and encouragement have made this journey extremely rewarding. Alongside Prof. Kollár, I want to thank my second reader, Prof. Zoltán Szabó, whose four-manifolds class this semester has been invaluable in helping me connect the algebraic and topological ideas throughout this thesis. Finally, thanks to Dhruv Goel, who has been extremely generous with his time over the past two semesters helping me learn more algebraic geometry.

Throughout Princeton, many professors have shaped my time here. A huge thanks to Jakub Witaszek, from whom I learned so much through his crash course in algebraic geometry in the Summer of 2025, and to Sung Gi Park, who's Riemann surfaces class I took my junior year started my interest in algebraic geometry. I also want to give particular thanks to Camillo De Lellis, Assaf Naor, Jacob Shapiro, Chris Skinner, Noga Alon, Lue Pan, and especially Alan Chang, who's enthusiasm in my introductory analysis class convinced me to study math in the first place.

My time at Princeton has been made so special by the amazing community built around the math department - not just math majors, but the people who spend their time around Fine Hall. While I can't name everyone, I want to shout out a few - Ananya Prasanna, Atharva Pathak, Brandon Cho, Colby Riley, Claire Kaneshiro, Daryna Yuschenko, Devan Shah, Elie Belkin, Greta Li, Julian Shah, Kaivalya Kulkarni, Kashti Umare, Kevin Xu, Linus So, Michael Kasprzak, Olivia Kwon, Ollie Thakar, and Shreya Sinha. Thanks for making my four years here a blast :)

Finally, I want to thank my family for their unwavering support throughout my entire life. First of all, thank you to my parents. They probably won't understand too much of this thesis, but they're proud of me nonetheless and they still let me try to explain it to them. Thanks to my two sisters Eva and Riana for always being there for me and keeping me sane during stressful times. And finally, thanks to my dog Pretzel, who definitely won't understand any of this thesis but does her job as a goofball that barks.

Declaration

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Signed, Kareem Jaber

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Chapter 1

Introduction

Here's an interesting question:

Question 1. What are all the compact complex surfaces up to biholomorphism?

This is an extremely hard question to answer, so hard that we don't have a complete answer to this day! In this thesis, we give a detailed exposition of this question's story when we restrict ourselves to the family of *complex projective surfaces*. By Chow's theorem, any complex projective manifold can be realized as a smooth algebraic variety, and every holomorphic map is defined by polynomials. Thus complex projective varieties and their classification carries an extremely rich structure that can be studied using both algebraic and transcendental methods. This thesis hopes to capture the interplay between complex and algebraic geometry, and motivate some of the powerful tools of algebraic geometry from the topological and holomorphic perspectives.

A general strategy to classification problems is to break it into two parts in the following way: if we are interested in classifying some family of objects under an equivalence relation $\sim_A$, then to attack this question, we may

1. Define a weaker notion of equivalence $\sim_B$, and instead classify our objects with respect to $\sim_B$ instead.

2. Then within each $\sim_B$ equivalence class, understand how objects (up to $\sim_A$ equivalence) are related to each other.

In the setting of complex projective surfaces, this question has been classically attacked with the weaker equivalence being *birational equivalence*: two algebraic varieties $X \to Y$ are said to be *birational* to each other if there are Zariski-open subsets $U \subset X$, $V \subset Y$ such that U is isomorphic to V. Then the problem of classifying complex projective surfaces breaks down into solving the following two steps:

1. Instead of classifying these surfaces up to (algebraic or holomorphic) isomorphism, we classify them up to birational equivalence.

2. Then within each birational equivalence class we understand the relationships between the individual complex surfaces.

For the second step of our motivating classification problem, it is of particular interest to find the "simplest" such complex surfaces in each birational equivalence class. This thesis is an exposition on constructing these simplest representatives, called the *minimal models* of complex surfaces, and understanding how the tools used to study them fit into the broader world of the birational geometry of higher dimensional algebraic varieties.

Before beginning this exploration, we take a moment to discuss what is known about the first step. The *Enriques-Kodaira* classification, carried out by Enriques in the early 1900s for complex projective surfaces, and extended to general complex surfaces by Kodaira in the 1960s, gives a rough classification of complex surfaces by the *Kodaira dimension*. For a complex manifold X, and for K_X the canonical class of the X (i.e. the divisor class corresponding to the determinant of the cotangent bundle of X) the Kodaira dimension $\kappa(X)$ measures the rate of growth of the dimension of global sections as we take tensor powers of K_X:

- $\kappa(X) = -\infty$ if $h^0(K_X^{\otimes n}) = 0$ for all n
- $\kappa(X) = k$ for $k \geq 0$ if $h^0(K_X^{\otimes n}) = O(n^k)$

It is a general fact that the growth rate of sections of tensor powers of any fixed line bundle $\mathcal{L}$ on a complex manifold X is bounded by $n^{\dim X}$. The Kodaira dimension is roughly a measure of the complexity of a variety. For example, for smooth curves, the Kodaira dimension of $\mathbb{P}^1$ is $-\infty$, the Kodaira dimension of any elliptic curve is 0, and the Kodaira dimension of any Riemann surface of genus two or greater is 1.

In the case of smooth surfaces, the Enriques-Kodaira classification gives a characterization of all minimal surfaces S, those that are the "simplest" in their birational equivalence class, as fitting inside one of the following subfamilies categorized by Kodaira dimension:

- **$\kappa(S) = -\infty$:** These are the simplest kinds of minimal surfaces. These consist of $\mathbb{P}^2$, $\mathbb{P}^1$ fibrations over a curve (known as geometrically ruled surfaces), and the family of Class VII surfaces (which are never algebraic surfaces).
- **$\kappa(S) = 0$:** These consist of complex tori and abelian varieties, the K3 surfaces, hyperelliptic surfaces (the product of two elliptic curves quotiented by a finite group of automorphisms), Enriques surfaces (all algebraic), and Kodaira surfaces (never algebraic).
- **$\kappa(S) = 1$:** All surfaces in this class are elliptic fibrations (though not all elliptic fibrations have Kodaira dimension 1), i.e. surfaces with a morphism to a curve that has smooth genus 1 curves fibers except at a finite number of points where the fibers have singularities.
- **$\kappa(S) = 2$:** These surfaces are known as *surfaces of general type*, and are the category we know the least about (and also by far the "largest" category). However, we do know that all surfaces of general type are algebraic, and that there are particular relations they must obey between their Chern classes.

Significant amounts of work has been done in the past several decades to better understand the finer structure within each of these subfamilies. For a much more complete treatment of the Enriques-Kodaira classification, see [Bea96] for the complex algebraic setting and [Bar+04] for the general setting.

1.1 An Overview of the Thesis

I've intended this thesis to be understandable to a student that has taken a one-semester course in algebraic geometry at the level of [Har77] or [Vak25], and is comfortable with the ideas of differential and algebraic topology. While almost everything in this thesis can be thought of as varieties, the language of sheaves and schemes will be frequently useful throughout various arguments. However, I have tried to motivate much of my arguments and add ample amounts of pictures, so I hope that those without the technical background can still enjoy a large part of this thesis. While the Enriques-Kodaira classification and the results stated in this thesis can be carried out over fields of any characteristic, we will always work over $\mathbb{C}$ and make use of transcendental techniques whenever we can[1].

In Chapter 2, we introduce arguably the most crucial tool in the study of algebraic surfaces - the intersection form. We define the intersection form both topologically and algebraically, and show their equivalence, allowing us to make use of both algebraic and transcendental techniques throughout the thesis. We also introduce the tools of Riemann-Roch and the adjunction formula, as well as some foundations in complex geometry.

Chapter 3 gives an exposition of many of the key ideas and results in the general structure of the birational geometry of surfaces. In particular, we show that any birational map between surfaces can be built from a composition of simpler birational maps called *blowups*. This allows us to define the notion of a minimal model of a surface S, in a sense a least complicated representative of the birational equivalence class of S, and we devote the rest of the paper to understanding such surfaces.

In Chapter 4, we find all the minimal models of ruled surfaces (those birational to $C \times \mathbb{P}^1$ for C a curve) and rational surfaces (those birational to $\mathbb{P}^2$), and understand the intersection theory of these surfaces. In these cases, we will get a very explicit characterization of these minimal models, but there will be many minimal models in a single birational equivalence class. In Chapter 5, we show that for any surface that is not ruled or rational, their birational equivalence class has a unique minimal model. While their minimal models can be significantly more complex, elements in their birational equivalence class are related in a much simpler way.

Finally, in Chapter 6, we unify many of the ideas and results of the previous chapters under the perspective of the Minimal Model Program, a framework developed in the 1980s built to help study the birational geometry of higher dimensional algebraic varieties. We present two foundational results of the program, highlighting how they fit within our previous study of minimal surfaces and how these tools can be extended past the surface case to threefolds and beyond. The Minimal Model Program and the birational geometry of high-dimensional algebraic varieties is a very rich and active area of mathematics to this day, and I hope this final chapter can provide a segway to further reading about the results and techniques in this area.

Chapters 3 through 5 loosely follow [Bea96] and [Vak02], and serve as helpful references while reading this paper. Both [Huy05] and [Bar+04] serve as great references for complex geometry as well.

[1] In fact, the transcendental techniques that we use still prove our statements over any characteristic zero field by the Lefschetz principle: roughly, any algebraic geometry statement over $\mathbb{C}$ also holds true over any algebraically closed field of characteristic zero.

1.2 Some Preliminaries and Assumptions

In this thesis, unless otherwise stated,

All surfaces are assumed to be smooth complex projective varieties.

For curves, we will be more careful in denoting when they are smooth or not.

As discussed before, our arguments will frequently jump between the complex analytic and complex algebraic categories. To treat all of our complex projective manifolds as varieties, we will be using Chow's Theorem:

> **Theorem 1.** (Chow's Theorem) Any complex projective manifold is in fact a complex algebraic variety. Furthermore, any holomorphic map between two complex projective manifolds is defined by polynomials, so is a morphism of the corresponding algebraic varieties.

More strongly, we will be using the fact that the cohomology of complex projective manifolds behave the same when working over either category. For this, we will be assuming Serre's *Géometrie Algébrique et Géométrie Analytique* [Ser56]:

> **Theorem 2.** (Serre's GAGA) The category of coherent algebraic sheaves is equivalent to the category of coherent analytic sheaves. Moreover, this correspondence preserves cohomology groups.

When considering line bundles on a surface S, say $\mathcal{L} = \mathcal{O}_S(D)$ with D a prime divisor for concreteness, we can view this line bundle as a subspace of our function field, or in this example as the space of functions that are regular on S up to a pole of order 1 along the divisor D. Serre's GAGA says that we can view these functions as either meromorphic or rational and the cohomology that we compute for these line bundles will be the same.

In working with smooth morphisms of complex projective varieties, the following result tells us that viewing our varieties as complex manifolds, smooth morphisms are just smooth submersions:

> **Lemma 1.** A morphism between smooth complex projective varieties $\varphi : X \to Y$ is smooth in the scheme-theoretic sense if and only if when we view X and Y as complex manifolds, the map is a smooth submersion, i.e. the differential is surjective on tangent spaces.

Similarly, we also have that the definitions of a proper morphism of complex projective varieties and proper maps of complex projective manifolds coincide (but this is an empty statement as every projective morphism is proper and every map between projective manifolds is proper). The purpose for smoothness conditions in this paper will be to guarantee that the fibers of our map are all homologous (when the codomain is connected) by Ehresmann's Lemma in differential topology:

> **Lemma 2.** (Ehresmann's Lemma) Let $\varphi : M \to N$ be a smooth map of smooth manifolds that is proper and a surjective submersion. Then φ is a locally trivial fibration.

In fact, smoothness of our morphism is stronger than what we need, and actually shows that all of our fibers are homologous through a family of diffeomorphisms. The correct level of regularity is *flatness*, but we will try to avoid using this language when possible.

We will be assuming Serre Duality, which we state here in the case of surfaces:

Theorem 3. (Serre Duality for surfaces) Let S be a smooth complex algebraic surface, ω_S the canonical bundle of S, and $\mathcal{L}$ a line bundle (equivalently, an invertible sheaf) on S. The cup product pairing

$$H^i(S, \mathcal{L}) \times H^{2-i}(S, \omega_S \otimes \mathcal{L}^*) \to \mathbb{C}$$

is a perfect pairing. In particular, we have that

$$H^i(S, \mathcal{L}) \cong H^{2-i}(S, \omega_S \otimes \mathcal{L}^*)^*$$

Finally, we will also need the result of Stein Factorization, which follows from Zariski's Connectedness Theorem:

Theorem 4. (Stein Factorization) Let $f : X \to Y$ be a proper morphism of varieties. Then f factors as

$$X \xrightarrow{f_1} Y' \xrightarrow{f_2} Y$$

where f_1 is surjective morphism with connected fibers and f_2 is a finite morphism.

Without further ado, we begin our study of complex projective surfaces!

Chapter 2

Topology and Algebraic Geometry

2.1 The Intersection Form

The study of surfaces introduces many new tools that don't arise in the study of curves. One particular tool is the intersection form, which describes how curves in a complex surface intersect. A complex projective surface S is both an algebraic variety and a complex manifold of dimension 2, so in particular inherits the structure of a four-dimensional compact oriented smooth manifold. The intersection form can be described from both these perspectives. Topologically, it is defined in the way one might expect, and algebraically, it is defined in a slightly more convoluted but more functional way. In this section, we discuss both perspectives and prove their equivalence for complex surfaces.

2.1.1 The Topological Intersection Form

We consider X to be a compact, connected, smooth, oriented four-manifold over the real numbers (strictly, we only need our manifold to be topological rather than smooth).

Definition 1. We define the *topological intersection form* $Q_X : H^2(X,\mathbb{Z}) \times H^2(X,\mathbb{Z}) \to \mathbb{Z}$ given by

$$Q_X(\alpha,\beta) = (\alpha \smile \beta, [X]).$$

Here, $[X]$ denotes the fundamental class of X. As X is an even dimensional manifold, the cup product is symmetric and Q_X is a symmetric bilinear form. Importantly, the cup product can be viewed dual to intersection in the following sense: if Y_1 and Y_2 are smoothly embedded oriented (real) surfaces in X, we can perturb them smoothly so that they intersect transversely. Then we have that

$$PD([Y_1]) \smile PD([Y_2]) = PD([Y_1 \cap Y_2]), \tag{2.1}$$

where PD denotes the Poincaré dual. See [Hut11] for a proof of (2.1) using the Thom class. As every second homology class in a four-manifold is represented by an smoothly embedded surface, the intersection form Q_X on any two cohomology elements can always be thought of this way. By the fact that Poincaré duality is a perfect pairing, if a homology class has zero intersection with all other homology classes, it must be a torsion homology class. From this, it is immediate that X is a unimodular form on the free part of $H^2(X, \mathbb{Z})$. A result of Freedman says that the homeomorphism type of a simply connected 4-manifold is essentially determined by it's intersection form. For more on the topological intersection form, see Chapter 1 of [Rob99].

2.1.2 The Algebraic Intersection Form

To define the intersection form algebraically, we consider C_1, C_2 to be distinct irreducible curves on a smooth projective surface S over $\mathbb{C}$, and take $x \in C_1 \cap C_2$. Consider the local ring $\mathcal{O}_{S,x}$ on S. Since $\mathcal{O}_{S,x}$ is a UFD, we get local equations f_1 of C_1 and f_2 of C_2. The *intersection multiplicity* of C and C' at x is defined to be the dimension of $\mathcal{O}_{S,x}/(f_1, f_2)$ as a complex vector space. The Nullstellensatz gives us that for k large enough, the kth power of the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{S,x}$ is contained in (f_1, f_2). By expanding any function of $\mathcal{O}_{S,x}$ into it's power series expansion, we see that $\mathcal{O}_{S,x}/\mathfrak{m}_x^k$ is a finite-dimensional $\mathbb{C}$-vector space, and so $\dim_{\mathbb{C}} \mathcal{O}_X/(f_1, f_2)$ is finite. With this, we define the algebraic intersection number:

Definition 2. For C_1, C_2 distinct irreducible curves, we define the *algebraic intersection number* $(C_1.C_2)_{alg}$ on curves C_1, C_2 as the sum of the intersection multiplicities at every point in $C \cap C'$. Two distinct irreducible curves C_1, C_2 in a complex projective surface meet at a finite set of points, so this number is finite.

As the intersection of two surfaces in the topological setting is only dependent on the homology classes of the surfaces, we expect that as we "wiggle" algebraic curves around, their intersection number should stay the same. An indicator that this should be possible is that given C, C' two curves, their intersection number is captured by data about line bundles, hinting that the intersection form should be defined on the Picard group rather than just curves or divisors, and that the intersection form should be invariant under linear equivalence. To see this, note that the intersection number of C and C' is exactly the dimension of global sections of the sheaf $\mathcal{O}_{C\cap C'}$ corresponding to the scheme theoretic intersection of C and C' in S. More precisely, $(C.C') = h^0(S, \mathcal{O}_{C\cap C'})$, where $\mathcal{O}_{C\cap C'} = \mathcal{O}_S/(\mathcal{I}_{C\cap C'})$. From this, it seems plausible to relate the intersection form of C and C' to some cohomological data of $\mathcal{O}_S(-C - C')$. This motivates a definition of the intersection form purely in terms of line bundles:

Definition 3. We define the intersection form on two line bundles $(\mathcal{L}, \mathcal{L})'_{alg} \in \mathrm{Pic}(S)$ as

$$(\mathcal{L}.\mathcal{L}')_{alg} = \chi(\mathcal{O}_S) - \chi(\mathcal{L}^*) - \chi(\mathcal{L}'^*) + \chi(\mathcal{L}^* \otimes \mathcal{L}'^*)$$

This intersection form is a symmetric bilinear form on $\mathrm{Pic}(S)$. For a proof, see I.6 and I.7 of [Bea96]. As we would hope, this intersection product on the Picard group agrees with our original intersection product on curves, and the following proof gives motivation for the more general definition.

Lemma 3. For C, C' distinct irreducible curves, then we have that

$$(\mathcal{O}_S(C).\mathcal{O}_S(C'))_{alg} = (C.C')$$

Proof. We first note that $\mathcal{O}_{C\cap C'}$ consists of the points $C \cap C'$, with the stalk at x being isomorphic to $\mathcal{O}_x(S)/(f, g)$. Then $C.C'$ defined for curves earlier is exactly equal to $h^0(S, \mathcal{O}_{C\cap C'})$. Given $s \in \mathcal{O}_S(C)$ and $s' \in \mathcal{O}_S(C')$ sections vanishing at C, C' respectively, this reduces the proof to showing that the sequence

$$0 \to \mathcal{O}_S(-C-C') \xrightarrow{a\mapsto(as',-as)} \mathcal{O}_S(-C) \oplus \mathcal{O}_S(-C') \xrightarrow{(b,c)\mapsto(bs+cs')} \mathcal{O}_S \to \mathcal{O}_{C\cap C'} \to 0$$

is exact, since the proof is finished after taking Euler characteristics. We can check exactness at stalks, so for f, g local equations for C, C', we're asking for the exactness of the sequence

$$0 \to \mathcal{O}_x \xrightarrow{a\mapsto(ag,-af)} \mathcal{O}_x \oplus \mathcal{O}_x \xrightarrow{(b,c)\mapsto(bf+cg)} \mathcal{O}_x \to \mathcal{O}_x/(f,g) \to 0$$

Exactness at first, third, and fourth terms of the sequence are immediate. At the second step, if $bf = -cg$, then by the fact that stalks on a complex surface are UFDs and f, g are coprime (they don't share an irreducible component), there exists a $d \in \mathcal{O}_x$ such that $b = dg$ and $c = -df$, showing exactness at this step. □

As the zeros of sections of a line bundle are linearly equivalent as divisors, Lemma 3 shows that the intersection number is invariant under linear equivalence of divisors. With this, given two curves C, C' intersecting at a point p, we can always find linearly equivalent divisors $\widetilde{C} \sim C, \widetilde{C'} \sim C'$ not necessarily effective, such that both $\widetilde{C}$ and $\widetilde{C'}$ don't pass through p but have the same intersection number as $C.C'$. This will be very useful in our study of complex surfaces.

Often we will abuse notation and use the algebraic intersection form to intersect curves with line bundles, with the understanding that the curve C is interpreted as the line bundle $\mathcal{O}_S(C)$. A helpful property that we will use throughout this section and the rest of the paper is the following:

Lemma 4. Let S be a smooth surface, C be a curve, and $\mathcal{L}$ a line bundle on S. Then

$$(C.\mathcal{L})_{alg} = \deg(\mathcal{L}|_C)$$

Proof. Let s be a section of $\mathcal{L}$ who's corresponding divisor D of zeros and poles do not contain C as a component (we can do this by moving every component of the zero and pole divisors of s away from a point of C). Let $\widetilde{s}$ denote the restriction of s to C. Now $\deg(\mathcal{L}|_C)$ can be computed by the degree of the divisor associated to $\widetilde{s}$, which can easily be seen to be exactly $C.D$ since if C and D intersect at a point $p \in S$, f is a local parametrization of C at p, and g is a local parametrization of a component of D, then

$$\dim_{\mathbb{C}} \mathcal{O}_{S,p}/(f,g) = \dim_{\mathbb{C}}(\mathcal{O}_{S,p}/f)/g = \dim_{\mathbb{C}} \mathcal{O}_{C,p}/g = \text{order of vanishing of } g|_C \text{ at } p$$

Since $(C.\mathcal{L})_{alg}$ is $(C.D)_{alg}$ by definition, we are done. □

2.1.3 Connecting the Two Definitions

To be able to use both topological and algebraic methods in the study of complex algebraic surfaces, it is necessary to show the equivalence of these two definitions. The bridge between these two different approaches is given by the *exponential exact sequence* of sheaves:

$$0 \to \mathbb{Z}_X \xrightarrow{\iota} \mathcal{O}_S \xrightarrow{e^{2\pi i}} \mathcal{O}_S^* \to 0$$

The first map ι is the obvious inclusion of sheaves, and the second map sends a function f to the nonvanishing function $e^{2\pi i f}$. The exactness of this sequence follows from the local existence of logarithms. We may consider the corresponding long exact sequence:

$$0 \to \mathbb{Z}_S \to O_S \to \mathbb{O}_S^* \to H^1(S, \mathbb{Z}_S) \to H^1(S, O_S) \to H^1(S, O_S^*) \to H^2(S, \mathbb{Z}_S) \to H^2(S, O_S) \to \dots$$

The space of line bundles $\mathrm{Pic}(S)$ is naturally identified with $H^1(S, O_S^*)$. The image of a line bundle $\mathcal{L} \in \mathrm{Pic}(S)$ under the connecting homomorphism $\mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$ associates to $\mathrm{Pic}(S)$ an element $c_1(\mathcal{L})$ of second cohomology known as the *first Chern class* of $\mathcal{L}$. The first Chern class of a line bundle of the form $O(D)$ has a topological interpretation that will aid us in answering our original question:

Theorem 5. Let C be a curve in S. Then after identifying the Čech cohomology group $H^2(S, \mathbb{Z}_X)$ with the singular cohomology group $H^2(S, \mathbb{Z})$, we have that $c_1(O_S(C)) = PD([C])$.

We postpone the proof of Theorem 5 briefly. This gives us a third definition of the intersection form:

Definition 4. For divisors D_1, D_2, we define the c_1-*intersection form* as

$$(D_1.D_2)_{c_1} = \langle c_1(O_S(D_1)) \smile c_1(O_S(D_2)), [S] \rangle$$

By Theorem 5, it is clear that

$$(D_1.D_2)_{c_1} = \langle c_1(O_S(D_1)), [D_2] \rangle = \langle [D_1], c_1(O_S(D_2)) \rangle$$

and that this definition is symmetric and bilinear. It is also clear by Theorem 5 that for curves C_1, C_2, the c_1-intersection form $(C_1.C_2)_{c_1}$ agrees with the topological intersection from $Q_X([C_1], [C_2])$.

Finally, we show that for curves in a surface S, the c_1-intersection form agrees with the algebraic intersection form. Note that every line bundle can be written as the difference of two very ample line bundles, and Bertini's theorem says that a generic section of a very ample line bundle has a *smooth* divisor of zeros. Then for any divisor D, we can write $O_S(D)$ as $O_S(D_1) - O_S(D_2)$ where D_1, D_2 are sums of smooth effective curves. Since both the c_1 and algebraic intersection form are bilinear, to show that $(D_1.D_2)_{c_1} = (D_1.D_2)_{alg}$, we may assume that D_1 is a smooth curve and that $O_S(D_2)$ has a section. We have the analogue of Lemma 4 for the c_1 intersection form:

$$(D_1.O_S(D_2))_{c_1} = \langle PD([D_1]) \smile PD([D_2]), [S] \rangle = \deg(O_S(D_2)|_{D_1})$$

where deg is now the "topological" degree of the line bundle, i.e. we perturb a global section of $O_S(D_2)$ restricted to D_1 so that it intersects the zero section transversely. Since the local ring of a smooth algebraic curve is a DVR, the statement we would like to show now boils down to saying that a function on a Riemann surface $C \to \mathbb{C}$ with a multiplicity k zero at a point p can be perturbed topologically so that the multiplicity k zero turns into k zeros of multiplicity 1. But this is clearly true, and with this we finally conclude that for two curves C_1, C_2,

$$Q_S([C_1], [C_2]) = (C_1.C_2)_{alg} = (C_1.C_2)_{c_1}$$

and we may drop the subscripts from our notation.

2.1.4 The c_1 Map and a Proof of Theorem 5

It remains to prove Theorem 5 to complete the proof of the agreement of all of our definitions of the intersection form. Why should we expect Theorem 5 to hold? That is, why should we expect this exact sequence to take a line bundle, and from it's transition functions extract information about the zeros of a generic section? The key idea is that the obstruction to finding global logarithms of a function is determined by how much the function winds around it's zeros. Given a smooth section s of a line bundle $\mathcal{L}$ on S, after removing the support of $D = \mathrm{div}(s)$ from S, we get a map

$$S - \mathrm{supp}(D) \xrightarrow{s} \mathbb{C} - 0$$

Now the existence of a global logarithm of such a section s by definition is the existence of such a map $\log s$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathbb{C} \arrow[d, "e^{2\pi i}"] \\ M - \mathrm{supp}(D) \arrow[ur, "\log(s)"] \arrow[r, "s"] & \mathbb{C} - 0 \end{tikzcd}$$

By covering space theory, this map exists if and only if the induced map on π_1 by s is trivial. The cohomology of the exponential exact sequence should capture the topological information of how much s winds around it's zero set, which by the argument principle corresponds to the multiplicity of the zero.

In order to prove the first Chern class of a line bundle is in fact the same thing as the zeros of a section of it, we need a way to explicitly relate Čech cohomology to singular/simplicial cohomology.

Definition 5. Given a open cover U_i of a topological manifold X, we define the *nerve* of the cover as the simplicial complex with vertices U_i, and k-cells given by

$$(i_{i_1}, \ldots, i_{i_k}) : U_{i_1} \cap \ldots \cap U_{i_k} \neq \emptyset$$

Leray's nerve theorem says that given a topological space X and a cover U_i such that all nonempty intersections are contractible, then the simplicial complex $N(X)$ given by the nerve of X is homotopy equivalent to X.

We prove Theorem 5 first for the case of Riemann surfaces.

Lemma 5. Let X be a Riemann surface. Then the map

$$\delta : H^1(X, \mathcal{O}_X^*) \to H^2(X, \mathbb{Z})$$

given in the long exact sequence corresponding to the exponential exact sequence sends the line bundle $\mathcal{O}(D)$ corresponding to an effective divisor D to $\deg(D)[X]$

Proof. By linearity of the maps above, it suffices to prove this statement for $D = p$ a point on X. The local neighborhood of p after removing p from X is $\mathbb{C}$ minus the origin. We will create a open cover U_i of X with the following properties:

- The U_is satisfy the property required of Leray's nerve theorem,
- The U_is give a trivialization of the line bundle $\mathcal{O}(p)$, and
- On every nonempty pairwise intersection $U_i \cap U_j$ the transition function g_{ij} of $\mathcal{O}(p)$ has a well-defined logarithm.

To do this, note that $\mathcal{O}(p)$ has trivializations given by $X - p$ and a small disk D around p lying in a complex chart. We refine this trivialization to a cover C such that D is covered by four charts $U_1, \ldots, U_4$ as in Figure 2.1, and the rest of X is covered by contractible open sets disjoint from D. As Čech cohomology can be computed on a cover with each open set homologically trivial, we compute the image of $\mathcal{O}(p)$ under the map $H^1(X, \mathcal{O}_X^*) \to H^2(X, \mathbb{Z})$ is defined as follows: given g_{ij} functions on $U_i \cap U_j$, find ℓ_{ij} such that $e^{2\pi i \ell_{ij}} = g_{ij}$. Since $g_{ij}g_{jk}g_{ki} = 1$ we must have $\ell_{ij} + \ell_{jk} + \ell_{ki} \in \mathbb{Z}$. The image of the 1-cocyle $(g_{ij}) \in H^1(X, \mathcal{O}_X^*)$ is the 2-cocycle in $H^2(X, \mathbb{Z})$ assigning the value $\ell_{ij} + \ell_{jk} + \ell_{ki}$ to the triple intersection $U_i \cap U_j \cap U_k$.

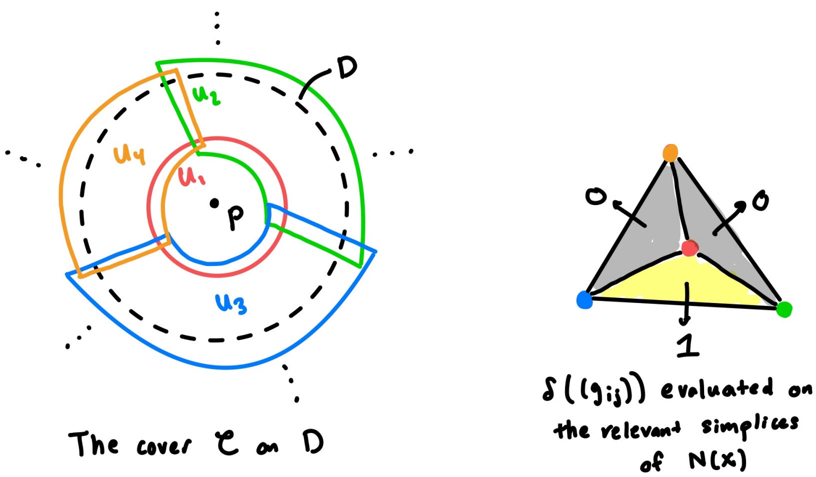

Figure 2.1: An illustration of the cover C and the evaluation of $\delta((g_{ij}))$ on $[X]$

Now note that the transition functions g_{ij} for $i > j$ are z in the local chart given by D for $i = 1, j = 2, 3, 4$, and 1 otherwise. On the chart D we pick branches of $\log(z)$ on $U_1 \cap U_2, U_1 \cap U_3, U_1 \cap U_4$, so that

- On $U_1 \cap U_2 \cap U_4$, the logarithms from $U_1 \cap U_2$ and from $U_1 \cap U_4$ agree
- On $U_1 \cap U_3 \cap U_4$, the logarithms from $U_1 \cap U_3$ and from $U_1 \cap U_4$ agree

However, as $\log z$ has a monodromy of $2\pi i$ as it goes around the origin, the logarithms of g_{13} and g_{23} will differ by $2\pi i$ on $U_1 \cap U_2 \cap U_3$. Therefore, we see that the Čech 2-cycle $\delta((g_{ij}))$ will assign the value 0 to $U_1 \cap U_2 \cap U_4$ and $U_1 \cap U_3 \cap U_4$, but will assign the value 1 to $U_1 \cap U_2 \cap U_3$; see Figure 2.1. On every other double intersection $U_i \cap U_j$, the transition function g_{ij} is 1, and we can choose the logarithm ℓ_{ij} of these g_{ij} to be 0.

Viewing $\delta((g_{ij}))$ as an element of the simplicial cohomology of the nerve $N(X)$ of X corresponding to the cover C, we see that $\langle \delta((g_{ij})), [X] \rangle = 1$, completing the proof of the lemma. □

While Theorem 5 can be proved by running another similar covering argument in $\mathbb{C}^2$, we will instead prove a more general version of 5 by bootstrapping the argument in the case of Riemann surfaces[1]:

Theorem 6. (More general Theorem 5) Let D by a divisor in X a smooth complex projective variety. Then $c_1(\mathcal{O}_S(D)) = PD([D])$

Proof. Since every line bundle is the difference of very ample line bundles, and by Bertini's theorem a general hyperplane section of a smooth variety is smooth, we can write $D = D_1 - D_2$ where D_1, D_2 are smooth and $\mathcal{O}_X(D_1)$, $\mathcal{O}_X(D_2)$ are very ample. Therefore, it suffices to prove the theorem for smooth divisors corresponding to very ample line bundles.

Note that Lemma 5 proves the theorem for the case of Riemann surfaces, and in particular, it is proven for $\mathbb{P}^1$. We will now prove the theorem for $\mathbb{P}^n$ for all n. Firstly, note that the c_1 map is functorial, as the long exact sequence of Čech cohomology is functorial (and a map between two varieties induces a map between their exponential sequences). The obvious embedding $f : \mathbb{P}^{n-1} \to \mathbb{P}^n$ is an isomorphism on the level of second cohomology (which can be seen via Poincaré duality), and therefore, we have the commutative diagram

$$\begin{tikzcd} \mathrm{Pic}(\mathbb{P}^n) \arrow[r, "c_1"] \arrow[d, "\wr"] & H^2(\mathbb{P}^n, \mathbb{Z}) \arrow[d, "\wr"] \\ \mathrm{Pic}(\mathbb{P}^{n-1}) \arrow[r, "c_1"'] & H^2(\mathbb{P}^{n-1}, \mathbb{Z}) \end{tikzcd}$$

Since the vertical maps are isomorphisms and f^* on H^2 sends the Poincaré dual of the hyperplane cohomology class of $\mathbb{P}^n$ to that of $\mathbb{P}^{n-1}$, we see by induction that for any n and H any hyperplane in $\mathbb{P}^n$ that

$$c_1(H) = PD([H])$$

Since $\mathrm{Pic}(\mathbb{P}^n)$ is generated by H, this proves the theorem for projective space.

We now prove the theorem for arbitrary smooth complex projective X, where we've reduced to studying smooth divisors D such that $\mathcal{O}_X(D)$ is very ample. Take the embedding corresponding to projective space corresponding to D, giving an embedding $\varphi : X \to \mathbb{P}^N$ and a hyperplane H_0 in $\mathbb{P}^N$ such that $H_0 \cap \varphi(X) = \varphi(D)$. Now again by functoriality of the c_1 map we have the commutative diagram

$$\begin{tikzcd} \mathrm{Pic}(\mathbb{P}^N) \arrow[r, "c_1"] \arrow[d] & H^2(\mathbb{P}^N, \mathbb{Z}) \arrow[d] \\ \mathrm{Pic}(X) \arrow[r, "c_1"'] & H^2(X, \mathbb{Z}) \end{tikzcd}$$

[1] If one chooses to go this route, you need to be careful about the smoothness of the divisor chosen. The reason why we were able to nicely make a cover in the Riemann surface case above was because in a neighborhood of a point, the complement of the point is isomorphic to S^1. Now if C is a smooth curve in a surface then topologically the complement of C in a neighborhood is just $\mathbb{C}^2 - \mathbb{C}$, and this gives you S^1 topologically as well. When C is singular, stranger things can happen. For instance, the complement of $\mathbb{C}^2 - \{xy = 0\}$ is now $S^1 \times S^1$. Even stranger behavior can happen at singular points - for an illustration of this, see for example Chapter 1 of [Mum76].

where the vertical maps are given by φ^*. Since φ^* on Pic sends H_0 to D, and sends $PD([H_0])$ to $PD([D])$ we get that

$$c_1(D) = PD([D])$$

This completes the proof of the theorem. □

2.1.5 The Intersection Pairing in Higher Dimensions

Theorem 6 gives us a way to define the intersection pairing on complex projective manifolds of any dimension! Instead of pairing curves with curves, we instead pair curves with divisors.

Now that X is no longer a surface, the group of (Weil) divisors are formal linear combinations of codimesion 1 subvarieties rather than just curves. To this end, we define the group of *1-cycles* Z_1 to be formal linear combinations of curves in X.

> **Definition 6.** For a general smooth complex projective variety X, we define the intersection form $Z_1(X) \times \mathrm{Div}(X) \to \mathbb{Z}$ for $D \in \mathrm{Div}(X)$, $Z \in Z_1(X)$ by
>
> $$(D.Z) := \langle c_1(D), [Z] \rangle = \langle c_1(D) \smile PD([Z]), [X] \rangle$$

By Theorem 6, $D.Z$ agrees with the topological intersection form on an n-dimensional manifold.

2.2 Which Homology Classes are Algebraic?

As we are interested in the interplay of complex and algebraic geometry, a particularly interesting question to ask is given a complex surface S, which homology classes have an algebraic representative? More precisely, *what second homology classes can be represented by algebraic 1-cycles, viewed as a real surface in the real 4-manifold S?* By algebraic 1-cycles, we mean formal $\mathbb{Z}$-linear combinations of curves lying inside a complex projective variety X. By Theorem 5 and Poincaré duality, this question is precisely asking for the image of $\mathrm{Pic}(S)$ in $H^2(S, \mathbb{Z})$ in the exponential exact sequence, which has a name:

> **Definition 7.** For a complex surface, the *Néron-Severi group $NS(S)$* is the image of $\delta : \mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$.

Using some basic Hodge Theory, we can get an even stronger understanding of this subgroup.

2.2.1 Some Background in Complex Geometry

Consider an arbitrary complex manifold X of dimension n. Over $\mathbb{R}$, we can interpret $H^k(X, \mathbb{R})$ as de Rham cohomology, the cohomology of the chain complex of smooth differential k-forms with the differential being the exterior derivative. Given a real vector space of dimension $2n$ with coordinates $(x_i, y_i)_{1 \le i \le n}$, functionals $T : \mathbb{R}^{2n} \to \mathbb{C}$ split into the sum of complex linear and complex antilinear functionals with respect to the complex variables $(x_i + iy_i)_{1 \le i \le n}$. In this way, a local basis of differential 1 forms $(dx_i, dy_i)_{1 \le i \le n}$ can instead be written in complex linear and complex anti-linear forms $(dz_i, d\overline{z}_i)_{1 \le i \le n}$.

Let $A^{p,q}$ be the subspace of differential $p+q$ forms generated by $dz_{i_1} \wedge \ldots \wedge dz_{i_p} \wedge d\overline{z}_{j_1} \wedge \ldots \wedge dz_{j_q}$, and define the differentials

$$\begin{aligned} \partial : A^{p,q} &\longrightarrow A^{p+1,q} & \partial &= \pi_{p+1,q} \circ d \\ \overline{\partial} : A^{p,q} &\longrightarrow A^{p,q+1} & \overline{\partial} &= \pi_{p,q+1} \circ d \end{aligned}$$

where $\pi_{p,q}$ denotes the projection onto the $A_{p,q}$ subspace of Ω^{p+q}. We define the *Dolbeault cohomology* $H^{p,q}(X)$ to be

$$H^{p,q}(X) = \frac{\ker(\overline{\partial} : A^{p,q} \longrightarrow A^{p,q+1})}{\operatorname{im}(\overline{\partial} : A^{p,q-1} \longrightarrow A^{p,q})}$$

While the space of smooth differential k-forms Ω^k decomposes into the direct sum $\bigoplus_{i+j=k} A^{p,q}$, it is not clear that this decomposition extends to cohomology - the differential d could "mix" the ∂ and $\overline{\partial}$ components. However, the Hodge Theorem tells us that this actually does happen:

Theorem 7. [The Hodge Theorem] Let X be a complex projective manifold. Then we have the decomposition on cohomology

$$H^k(S, \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(X)$$

Moreover, under complex conjugation we have that

$$\overline{H^{p,q}}(X) \cong H^{q,p}(X)$$

We can relate de Rham and Dolbeault cohomology to sheaf cohomology through the language of derived functor cohomology. Given a sheaf $\mathcal{F}$, we consider an injective resolution, i.e. an exact sequence of sheaves of the form

$$0 \longrightarrow \mathcal{F} \longrightarrow I_0 \longrightarrow I_1 \longrightarrow I_2 \longrightarrow \ldots$$

where the I_is are injective sheaves, meaning than any map from a sheaf A to I can be extended to a map $B \longrightarrow I$ for any sheaf B containing A. After applying global sections, we get a complex

$$0 \longrightarrow \Gamma(X, I_0) \longrightarrow \Gamma(X, I_1) \longrightarrow \Gamma(X, I_2) \longrightarrow \ldots$$

The derived functor cohomology of $\mathcal{F}$ is given by the cohomology of the above complex. It can be shown that the derived functor cohomology of $\mathcal{F}$ can be computed using an *acyclic* resolution of sheaves, meaning that each sheaf in the resolution has vanishing H^i for $i \geq 1$, and that the cohomology is independent of choice of injective/acyclic resolution.

From this perspective, we can define the cohomology of a complex surface S with respect to O_S by considering a resolution by sheaves

$$0 \longrightarrow O_S \longrightarrow A^{0,0} \longrightarrow A^{0,1} \longrightarrow A^{0,2} \longrightarrow 0 \tag{2.2}$$

where $A^{i,j}$ denotes the space of (i,j)-forms, and the differentials are given by $\overline{\partial}$. The exactness of this resolution follows from the $\overline{\partial}$ Poincaré-lemma. As the higher cohomology of the sheaf $A^{i,j}$ on S vanishes (in particular, the sheaf $A^{i,j}$ admits partitions of unity), we see that the cohomology $H^i(X, \mathcal{O}_S)$ is given exactly by the Dolbeault cohomology $H^{0,i}(S)$.

More generally, one can show the Dolbeault Theorem:

Lemma 6. (The Dolbeault Theorem) For any complex projective manifold, we have that $H^{i,j}(X) = H^j(X, \Omega^i_X)$, where Ω^i_X is the space of holomorphic differential i forms.

2.2.2 The Lefschetz 1-1 Theorem

With the tools of Hodge theory, we can exactly describe how homology classes with algebraic representatives sit inside $H^2(X, \mathbb{Z})$, giving us a characterization of the Néron-Severi group. We start with the following simple lemma:

Lemma 7. If X is a complex projective manifold, consider the map $H^k(X, \mathbb{C}_X) \to H^k(X, \mathcal{O}_X)$ induced by the injection of sheaves $\mathbb{C}_X \to \mathcal{O}_X$. Viewing $H^k(X, \mathbb{C})$ as de Rham cohomology and $H^k(X, \mathcal{O}_X)$ as Dolbeault cohomology, the map $H^k(X, \mathbb{C}_X) \to H^k(X, \mathcal{O}_X)$ is given by projection onto the $(0, k)$ component.

Proof. Given the standard resolutions of $\mathbb{C}_X$ and $\mathcal{O}_X$, we have a map of resolutions given by projecting to the $(0, i)$ forms:

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{C}_X \arrow[r] \arrow[d, "j"'] & \Omega^0(X) \arrow[r, "d"] \arrow[d] & \Omega^1(X) \arrow[r, "d"] \arrow[d] & \Omega^2(X) \arrow[r] \arrow[d] & \cdots \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] & A^{0,0} \arrow[r, "\overline{\partial}"] & A^{0,1} \arrow[r, "\overline{\partial}"] & A^{0,2} \arrow[r] & 0
\end{tikzcd}$$

The Lemma now follows immediately from the Hodge decomposition. □

Now we can characterize the image of $\mathrm{Pic}(S)$ under the connecting homomorphism - more precisely, we will exactly compute the image of $NS(S)$ under the map $H^2(S, \mathbb{Z}_S) \to H^2(S, \mathbb{C}_S)$. Note that the image of $\mathrm{Pic}(S)$ inside $H^2(S, \mathbb{Z}_S)$ is the same as the kernel of the map $H^2(S, \mathbb{Z}_S) \to H^2(S, \mathcal{O}_S)$. This map factors through $H^2(S, \mathbb{C}_S)$ (since the map of sheaves $\mathbb{Z}_S \to \mathcal{O}_S$ clearly factors through $\mathbb{C}_S$):

$$\begin{tikzcd}
\mathrm{Pic}(S) \arrow[r, "\delta"] & H^2(S, \mathbb{Z}_S) \arrow[r, "\iota"] \arrow[d, "i^*"] & H^2(S, \mathcal{O}_S) \\
& H^2(S, \mathbb{C}_S) \arrow[ur, "j^*"'] &
\end{tikzcd}$$

The image of $\mathrm{Pic}(S)$ under δ is exactly the kernel of ι, thus the image of $\mathrm{Pic}(S)$ under $i^* \circ \delta$ is the image of i^* intersected with the kernel of j^*. The map j^* is projection on the $(0, 2)$ part by the previous lemma, therefore the subgroup of $H^2(S, \mathbb{C})$ we seek is the intersection of the image of $H^2(S, \mathbb{Z})$ with the $(2, 0)$ and $(1, 1)$ summands of $H^2(S, \mathbb{C})$. However the image of $H^2(S, \mathbb{Z})$ inside $H^2(S, \mathbb{C})$ is real, and so equal to it's conjugate. Therefore, we conclude that

Theorem 8. (Lefschetz's (1-1) Therorem) Let S be a complex projective surface and $i : \mathbb{Z}_S \to \mathbb{C}_S$ be the inclusion of sheaves. Then we have that

$$i^*(NS(S)) = i^*(H^2(S,\mathbb{Z})) \cap H^{1,1}(S)$$

From this point of view, we can highlight the "rigidity" of the cohomology classes with algebraic representatives. Poincaré duality says that for a real four manifold X, a smoothly embedded surface Y representing a homology class of X is uniquely determined up to torsion by its intersection with all other smoothly embedded surfaces in X. The Lefschetz (1-1) theorem tells us that for complex curves lying inside a complex surface S, their homology classes are uniquely determined up to torsion by *intersections with other complex curves* - so in some sense, this topological data of algebraic object (in this case, the Poincaré dual of a curve) can be defined almost entirely algebraically!

Precisely, we define two divisors D_1, D_2 in a complex surface S to be *numerically equivalent* if for all irreducible curves C,

$$D_1.C = D_2.C$$

The Lefschetz (1-1) theorem gives us the following corollary:

Corollary 1. On a complex surface S divisors D_1, D_2 are numerically equivalent if and only if $[D_1], [D_2] \in H^2(S,\mathbb{Z})$ are equal up to torsion.

Proof. If a divisor D is homologically trivial, then Poincaré duality and the fact that the topological intersection form agrees with the algebraic intersection form gives immediately that D is numerically equivalent to zero. Now for the other direction, suppose a divisor D is numerically trivial, so by the Lefschetz (1-1) theorem the fundamental class of D evaluates to zero on all elements of $i^*(H^2(X,\mathbb{Z})) \cap H^{1,1}(X,\mathbb{C})$, thus all of $H^{1,1}(X,\mathbb{C})$ since $i^*(H^2(X,\mathbb{Z}))$ is a lattice in $H^2(X,\mathbb{C})$. Now consider the pairing of the fundamental class of D with a cohomology element α lying entirely in $H^{0,2}(X,\mathbb{C}) \cup H^{2,0}(X,\mathbb{C})$. Then we have that

$$\langle [D], \alpha \rangle = \langle PD([D]) \smile \alpha, [S] \rangle = \langle c_1(D) \smile \alpha, [S] \rangle$$

Viewing $c_1(D)$ and α as an element of de Rham cohomology, $c_1(D) \smile \alpha$ is the wedge product between a $(1,1)$ form and a linear combination of $(2,0)$ and $(0,2)$ forms. Clearly this wedge product is zero. From this, we see that the fundamental class of D against *any* element of $H^2(S,\mathbb{C})$ is zero, and by Poincaré duality, D must be a torsion element of $H^2(S,\mathbb{Z})$. □

While the concept of divisors being homologous only makes sense over a topological space, the concept of numerical equivalence can be utilized over any base field. In this way, numerical equivalence can be viewed as a sort of generalization of homological equivalence in algebraic geometry.

The concept of linear equivalence of divisors is much further removed from homological equivalence of divisors. To see this, we look at the kernel of the c_1 map:

Definition 8. We define Pic^0 to be the kernel of the c_1 map $\mathrm{Pic}(X) \to H^2(S, \mathbb{Z})$ in the exponential exact sequence.

We will show that Pic^0 is often nonempty and also highly structured. Since X is compact, $H^0(X, \mathcal{O}_X^*) = \mathbb{C}^*$, and since $H^1(X, \mathbb{Z})$ is a finitely generated group the the image of $H^0(X, \mathcal{O}_X^*) \to H^1(X, \mathbb{Z})$ is trivial, thus $H^1(X, \mathbb{Z}) \to H^1(X, \mathcal{O}_X)$ is injective. Therefore, from the exactness of the exponential exact sequence, we have that

$$\mathrm{Pic}^0 = H^1(X, \mathcal{O}_X)/H^1(X, \mathbb{Z})$$

With this, we show the following simple result, who's proof parallels that of the Lefschetz 1-1 theorem:

Lemma 8. If X is a complex projective manifold, then Pic^0 is a torus of dimension $b_1(X)$.

Proof. The map $\varphi : H^1(X, \mathbb{Z}) \to H^1(X, \mathcal{O}_X)$ factors as

$$H^1(X, \mathbb{Z}_X) \to H^1(X, \mathbb{R}_X) \to H^1(X, \mathbb{C}_X) \to H^1(X, \mathcal{O}_X)$$

where the first two maps are obvious, and the third map $H^1(X, \mathbb{C}) \to H^1(X, \mathcal{O}_X)$ is projection onto the $H^{0,1}$ component of the Hodge decomposition by Lemma 7. Consider the map $H^1(X, \mathbb{R}) \to H^{0,1}(X)$. If the image of some cohomology class α is zero, then since α is a real cohomology class the image of it's conjugate is also zero, so α is zero in $H^{1,0}$ as well. Therefore, α is the zero cohomology class in $H^1(X, \mathbb{C})$ and thus also in $H^1(X, \mathbb{R})$, and so the map $H^1(X, \mathbb{R}) \to H^1(X, \mathcal{O}_X)$ is injective. These spaces both have the same rank as $\mathbb{R}$-vector spaces (they are rank $b_1(X)$), so the map is an isomorphism. Since the image of $H^1(X, \mathbb{Z})$ in $H^1(X, \mathbb{R})$ is a lattice, $H^1(X, \mathcal{O}_X)/H^1(X, \mathbb{Z})$ is a complex torus of dimension $b_1(X)$, completing the proof. □

Therefore, when $b_1(X)$ is nonzero (for example, X is a geometrically ruled surface over a curve of positive genus - see the later section on ruled and rational surfaces), linear equivalence is a strictly finer equivalence than homological and numerical equivalence.

2.3 The Normal Bundle, Adjunction, and Riemann Roch

For a smooth curve C sitting inside a surface S (more generally, a codimension 1 subvariety/submanifold Y of a complex variety/manifold X), the intersection form gives us a way to relate the canonical bundle of S to the canonical bundle of C. Holomorphically, the normal bundle of Y in X is defined to be the line bundle fitting into the exact sequence

$$0 \to \mathcal{T}_Y \to \mathcal{T}_X|_Y \to \mathcal{N}_{Y/X} \to 0 \tag{2.3}$$

For a general short exact sequence of vector bundles $0 \to E \to F \to G \to 0$, we get that $\det(F) = \det(E) \otimes \det(G)$, so we get from (2.3) that

$$\omega_Y = \omega_X|_Y \otimes \det(\mathcal{N}_{Y/X}). \tag{2.4}$$

Algebraically, if we have a closed embedding of smooth varieties $i : Y \to X$, and $\mathcal{I}_Y$ the ideal sheaf of Y in X, the conormal sheaf $\mathcal{N}^*_{Y/X}$ is defined to be $i^*(\mathcal{I}_Y/\mathcal{I}_Y^2)$. To understand this definition, note that the conormal bundle fits into the dual of the exact sequence (2.3),

$$0 \to \mathcal{N}^*_{Y/X} \to i^*(\Omega(X)) \to \Omega(Y) \to 0, \tag{2.5}$$

where Ω denotes the cotangent sheaf. At a point p of a variety X, the cotangent space is just $\mathfrak{m}_p/\mathfrak{m}_p^2$ for $\mathfrak{m}_p$ the maximal ideal of the local ring $\mathcal{O}_{X,x}$. The kernel of the map $i^*(\Omega(X)) \to \Omega(Y)$ on the stalk of a point p are exactly the elements in $\mathfrak{m}_p/\mathfrak{m}_p^2$ represented by functions that vanish along Y, which is clearly identified with the stalk of $\mathcal{I}_Y/\mathcal{I}_Y^2$, which proves that (2.5) is exact. By the same general fact as before about taking determinants of exact sequences of vector bundles, now applied to locally free sheaves, we get (2.4) in the algebraic setting.

If we choose Y to be a smooth curve C in and X to be a smooth surface S, then the conormal bundle $\mathcal{N}^*_{Y/X}$ is just the restriction of $\mathcal{O}_S(-C)$ to the curve C, denoted $\mathcal{O}_C(-C)$. To see this, first note that $\mathcal{I}_Y/\mathcal{I}_Y^2$ is locally free of rank 1, which we can easily check on stalks. To check that $\mathcal{I}_Y/\mathcal{I}_Y^2$ has the same transition functions as $\mathcal{O}_C(-C)$, let f_i be functions locally representing C in trivializing neighborhoods U_i of $\mathcal{I}_Y/\mathcal{I}_Y^2$. If $gf_i = g'f_j$ on $U_i \cap U_j$, then $g' = (f_i/f_j)g$, exactly the transition functions of $\mathcal{O}_C(-C)$. Therefore, we see that $\mathcal{N}_{Y/X} = \mathcal{O}_C(C)$.

Plugging this knowledge into (2.4) gives us that

$$\omega_C = \omega_S|_C \otimes \mathcal{O}_C(C)$$

Now we take degrees of the line bundles on both sides. On the left, as C is a smooth curve, we get $2g(C) - 2$, where $g(C)$ is the genus of the curve. On the right side, by Lemma 4, we get

$$\deg(\omega_S|_C \otimes \mathcal{O}_C(C)) = \deg(\omega_S|_C \otimes \mathcal{O}_S(C)|_C) = K_S.C + C^2$$

Therefore, we get

Corollary 2. (The Adjunction Formula, also known as the Genus Formula) For C a smooth curve in a smooth surface S, we have that

$$2g(C) - 2 = K_S.C + C^2$$

Before continuing, we make two quick notes about the normal bundle.

Understanding self-intersections: Topologically, the self-intersection of a complex algebraic curve in a complex algebraic surface has a geometric interpretation. We can identify the (topological) normal bundle of our complex curve C(viewed as a real surface) as a tubular neighborhood of it. A deformation of this complex curve off of itself that intersects the original curve transversely gives a section of the normal bundle, and the intersection number of the deformation with the original curve gives the self-intersection number of C.

This topological intuition carries into the algebraic world as well, since the degree of the normal bundle

$\mathcal{O}_C(C)$ of a smooth curve C in S is exactly C^2 by Lemma 4. If $C^2 = n$ for $n > 0$, then what this is saying is that if we "perturb C" to some linearly equivalent effective divisor, it intersects C with total multiplicity n. If $C^2 = 0$, then what we're saying is that we can only algebraically perturb C "parallel to itself" in S. In the case that C has negative self-intersection number, while C can be perturbed topologically to intersect itself with it's orientation reversed, C is algebraically "stuck" (i.e. it has no linearly equivalent effective divisors, since distinct algebraic curves must intersect each other nonnegatively). Choosing a non-effective divisor D linearly equivalent to C and not containing C as a common component, we need to delete the points of C where the negative parts of D intersect it before being able to perturb C algebraically.

The residue exact sequence for surfaces: On the topic of the normal bundle, we will often come across exact sequences of the form

$$0 \to \mathcal{O}_S \to \mathcal{O}_S(C) \to \mathcal{O}_C(C) \to 0$$

This can be viewed as the analogue of the "residue exact sequence for Riemann surfaces"

$$0 \to \mathcal{O}_C \to \mathcal{O}_C(p) \to \mathbb{C}_p \to 0$$

The map $\mathcal{O}_S \to \mathcal{O}_S(C)$ is defined by sending $1 \in \mathcal{O}_S$ to the canonical global section of $\mathcal{O}_S(C)$[2]. The map $\mathcal{O}_S(C) \to \mathcal{O}_C(C)$ is defined locally: for f a local parametrization of C at a point x, the map $\mathcal{O}_S(C)_x \to \mathcal{O}_C(C)_x$ sends g to $g \bmod f$. One helpful way of understand this map is by considering $\mathcal{O}_S(C)$ as embedded in the function field of S, then we can view a function g in the stalk of $\mathcal{O}_S(C)$ at x as the rational function $\frac{g}{f}$, and the map sends this rational function to it's "residue with respect to f". Note that this image lies in $\mathcal{O}_C(C)$ rather than just $\mathcal{O}_C$ because the local parametrization of C changes on different trivializations of the line bundle. This captures the fact that the reside with respect to f lies in the normal direction of C.

2.3.1 Generalizing the Genus Formula

In it's current state, the genus formula only holds for smooth curves. We may extend this result to singular curves using the Riemann-Roch formula for surfaces.

> **Theorem 9.** (Riemann-Roch for surfaces) For S a smooth complex algebraic surface and $\mathcal{L} \in \mathrm{Pic}(S)$, we have that
>
> $$\chi(\mathcal{L}) = \chi(\mathcal{O}_S) + \frac{1}{2}(\mathcal{L}^2 - \mathcal{L}.\omega_S)$$

Proof. By the bilinearity of the intersection form we can write the desired equation as

$$2\chi(\mathcal{O}_S) - 2\chi(\mathcal{L}) \stackrel{?}{=} \mathcal{L}^{-1}.(\mathcal{L} \otimes \omega_S^{-1})$$

[2] More explicitly, let C be locally cut out by f_i on the trivializations U_i. Then the line bundle $\mathcal{O}_S(C)$ has transition functions φ_{ji} from U_i to U_j given by $\frac{f_j}{f_i}$. This gives us a canonically defined global section of $\mathcal{O}_S(C)$ given by choosing the function f_i on each trivialization U_i. After embedding $\mathcal{O}_S(C)$ into the function field of S, we see that this global section corresponds to the constant function 1 on S, so embedding both $\mathcal{O}_S$ and $\mathcal{O}_S(C)$ into the function field of S this map is just the obvious inclusion.

But by the algebraic definition 3 of the intersection form, we have that

$$\mathcal{L}^{-1}.(\mathcal{L}\otimes\omega_S^{-1}) = \chi(\mathcal{O}_S) - \chi(\mathcal{L}) - \chi(\mathcal{L}^{-1}\otimes\omega_S) + \chi(\omega_S)$$

By Serre duality, $\chi(\mathcal{L}^{-1}\otimes\omega_S) = \chi(\mathcal{L})$ and $\chi(\omega_S) = \chi(\mathcal{O}_S)$, which immediately completes the proof of the theorem. □

With this, we generalize the genus formula to the case of singular curves.

Corollary 3. (General genus formula) Let C be an irreducible (not necessarily smooth) curve, and $p_a(C) = h^1(C, \mathcal{O}_C)$ it's arithmetic genus (which agrees with the topological genus when C is smooth). Then we have the genus formula

$$2p_a(C) - 2 = C^2 + C.K_S$$

Proof. We consider the residue exact sequence

$$0 \to \mathcal{O}_S(-C) \to \mathcal{O}_S \to \mathcal{O}_C \to 0$$

and take Euler characteristics. This gives us that

$$1 - p_a(C) + \chi(\mathcal{O}_S(-C)) = \chi(\mathcal{O}_S)$$

Replacing $\chi(\mathcal{O}_S(-C))$ with the expression from Riemann-Roch and rearranging gives the desired result. □

Remark . While we will mostly only be using the genus formula for smooth curves, the genus formula for singular curves will be useful for us in a particular way. Suppose that we have an irreducible curve C in a smooth surface S, and we are able to show that $C^2 + C.K_S = -2$, so that $p_a(C) = 0$. Then since taking the normalization N of a singular curve C must strictly decrease the arithmetic genus of the curve, yet $p_a(N)$ must be a nonnegative number, we see that C must be smooth with $p_a(C) = 0$, thus is rational.

Chapter 3

The Birational Geometry of Surfaces

3.1 A Quick Overview on Line Bundles and Maps to Projective Space

Given a line bundle $\mathcal{L}$ on a variety X, we can consider it's space of sections $H^0(X, \mathcal{L})$. From such a line bundle, we can construct a map to projective space as follows: take a basis $s_0, \dots, s_n$ of $H^0(X, \mathcal{L})$, and define the rational map

$$\varphi : X \dashrightarrow \mathbb{P}^n \qquad \varphi(x) = [s_0(x) : s_1(x) : \cdots : s_n(x)]$$

This map is rational as it is not defined where the s_i are all zero. If $\mathcal{L}$ is *basepoint free*, meaning that the global sections $H^0(X, \mathcal{L})$ don't all share a common zero, then this rational map is a morphism. Changing the basis of our space of sections only changes our image by an automorphism of $\mathbb{P}^n$. We can also do this same construction with a subspace of $H^0(X, \mathcal{L})$.

We may think about this map in a coordinate-free way. We recall that we have a correspondence

$$\left\{ \begin{array}{c} \text{Effective divisors linearly} \\ \text{equivalent to } D \end{array} \right\} \Longleftrightarrow \left\{ \begin{array}{c} \text{Nonzero sections of} \\ \mathcal{O}_S(D) \text{ mod scalars} \end{array} \right\},$$

To any effective divisor D, there is a global section s of $\mathcal{O}_S(D)$ who's divisor of zeros is precisely D. Conversely, given any global section s of $\mathcal{O}_S(D)$, the divisor of zeros $Z(s)$ is linearly equivalent to D and $\mathcal{O}_S(D) \cong \mathcal{O}_S(Z(s))$. Furthermore, any two sections with the same divisor of zeros locus differ by a scalar. For more details, see Chapter 2.3 of [Huy05]. Now let $|D|$ be the space of all effective divisors on S linearly equivalent to D. Under the above correspondence, we have a natural identification of $|D|$ with $P = \mathbb{P}(H^0(X, \mathcal{O}_X(D)))$. We identify a point of P^* (the dual of P) as a hyperplane in P. Then from this viewpoint, the map $\varphi : X \dashrightarrow \mathbb{P}^n$ can be defined as follows. Take a point $x \in X$ not a basepoint of $|D|$. As the linear functional on P of the form $s \mapsto s(x)$ is nontrivial, it's kernel is the $\dim(|D|) - 1$ dimensional subspace consisting of all divisors passing through x. We map x to the point in P^* corresponding to this hyperplane in P. It is easy to check that this description of φ is the same as the previous description. As before, φ can also be defined for a subspace V of $|D|$ analogously to this discussion. We call a subspace V of $|D|$ a *linear system*, and the space $|D|$ a *complete linear system*. The dimension of V and $|D|$ is the dimension of the projective space it defines.

From here, we see that the map φ induced by the linear system $V \subset |D|$ is injective if and only if for every two points $x, y \in V$ there exists an effective divisor $D' \in V$ such that one of x, y passes through D but the other doesn't. The injective map determined by φ is an embedding (equivalently, $\mathcal{O}_X(D)$ is *very ample*), if and only if for every x, the cotangent space at x is spanned by the differentials of local functions of divisors $D \subset V$ passing through x[1]. For X a complex surface, this is equivalent to asking for the tangent directions of the curves defined by the divisors in V to span the tangent space of X at x.

Under the map $\varphi : X \to \mathbb{P}^n$ induced by $|D|$, pullbacks of hyperplanes in $\mathbb{P}^n$ are exactly the divisors $|D|$. To see this, consider a hyperplane cut out by $a_0X_0 + \ldots + a_nX_n$ in $\mathbb{P}^n$. Now the pullback is the zero set of the section $a_0s_0 + a_1s_1 + \ldots + a_ns_n$, hence a divisor in $|D|$. In other words, $\varphi^*(\mathcal{O}_{\mathbb{P}^n}(1)) = \mathcal{O}_X(D)$.

Lastly, we consider the special case of *pencils* on a surface, i.e. a linear system of dimension 1, which will frequently arise in our study of algebraic surfaces. Consider a 2-dimensional subspace H of a complete linear system $|D|$, generated by effective divisors C_0 and C_1 (more precisely, by the sections of $\mathcal{O}_X(D)$ they determine). Away from the basepoints of the family generated by these two divisors, we get a map $\varphi : X \to \mathbb{P}^1$. A fiber of a point $[a : b] \in \mathbb{P}^1$ is the zero set of the section $bs_0 - as_1 \in \mathcal{O}_S(D)$, and thus the fibers of φ correspond exactly to the divisors in the subspace spanned by C_0 and C_1.Therefore, a pencil is some kind of "fibration" over $\mathbb{P}^1$, with fibers the divisors in our linear system, as depicted in Figure 3.1.

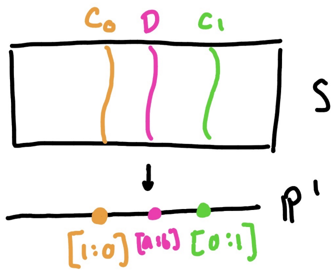

Figure 3.1: A pencil on a surface

3.2 What Can Go Wrong Extending Rational Maps of Surfaces?

Our main goal is to understand "all smooth complex projective manifolds of dimension 2" up to biholomorphism, or equivalently, "all projective algebraic varieties of dimension 2" up to biregular morphisms. As discussed in the introduction, this problem will be approached by instead studying surfaces up to birational morphism, so we better understand what can obstruct rational maps of surfaces from extending to an actual morphism.

We start with a warm up for smooth curves. Any rational map between smooth curves extends uniquely to a regular map by the following lemma:

> **Lemma 9.** For any rational map φ between smooth projective varieties $X \dashrightarrow Y$, this map extends uniquely to a $\widetilde{\varphi}$ defined everywhere but a subset of codimension at least 2.

[1]To see this, pull back the generators of the cotangent space at $\varphi(x) \in \mathbb{P}^n$.

Proof. We may immediately reduce to the case that $Y = \mathbb{P}^n$. Now the rational map $\varphi : X \dashrightarrow \mathbb{P}^n$ is of the form $[\varphi_0 : \varphi_1 : \cdots : \varphi_n]$, where φ is undefined where the φ_i have common zeros. Locally in X, we may scale the φ_i so that their greatest common factor is 1 (using the fact that X is smooth so it's local rings are UFDs), and thus their zero sets have no common components. Then the locus at which φ is not defined is the intersection of the zero sets of φ_i, hence the intersection of codimension 1 sets not sharing any component, hence codimension 2. $\square$

Therefore for smooth curves, a regular map defined on a dense open subset always extends to a regular map on the whole curve, and classification up to isomorphism is the same as birational classification.

For smooth surfaces, there may be obstructions to extending a map defined away from a point to the whole surface. To see this, we consider an example. We try to construct a map from $\mathbb{P}^1 \times \mathbb{P}^1$, viewed as the subset $V = \{xy - zw = 0\} \subset \mathbb{P}^3$ via the Segre embedding, to $\mathbb{P}^2$ as follows. First, fix a point of p of V, and pick a plane $L \subset \mathbb{P}^3$ disjoint from p such that in a chart containing p, L is parallel to the tangent plane at p (this is not necessary for the map, but helpful for the picture in Figure 3.2). For a point $q \neq p$ on V, take the line ℓ through p and q. Now ℓ intersects L in a point, call it $\varphi(q)$. The map $q \mapsto \varphi(q)$ defines a regular map from $V - p \to \mathbb{P}^2$, but doesn't extend to a regular map on all of V. As we limit q toward p, the limiting value of the $\varphi(q)$ precisely depends on the local direction at which we approach p. To be able to allow φ to be a regular map on all of V, we need to modify V in a way that "separates lines through p." We need a way to geometrically modify V so that p is replaced with an object that "remembers what direction it was approached from."

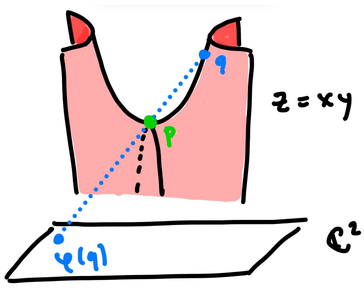

Figure 3.2: A visualization of $\varphi : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^2$ in the chart $\{[x : y : z : w] \in \mathbb{P}^3 : w \neq 0\}$

We build a local model of such a construction as follows: Consider the tautological line bundle $O(-1)$ of $\mathbb{P}^1$ given by the subset of points $(p, [\ell]) \in \mathbb{C}^2 \times \mathbb{P}^1$ such that p is in ℓ (the line represented by the point $[\ell] \in \mathbb{P}^1$). The projection π from $O(-1)$ to $\mathbb{P}^1$ is just the standard line bundle projection map. The other projection ε, from $O(-1)$ to $\mathbb{C}^2$, is what will be our local model for a blowup of a surface. Note that the fibers of ε are a point for every $p \in \mathbb{C}^2 - 0$, and moreover ε defines an isomorphism from $O(-1) - \varepsilon^{-1}(0)$ to $\mathbb{C}^2 - 0$. The fiber of 0 is a $\mathbb{P}^1$, and approaching the origin of $\mathbb{C}^2$ from different directions corresponds to approaching different points of the $\mathbb{P}^1$ corresponding to the fiber of 0 in the blowup.

To visualize this, we consider the picture over $\mathbb{R}^2$. Note that $O(-1) \subset \mathbb{R}^2 \times \mathbb{RP}^1$ over $\mathbb{RP}^1$ is the Möbius strip. The projection onto the $\mathbb{P}^1$ is projection onto the core circle, and the projection ε onto $\mathbb{R}^2$ can be seen in Figure 3.3. Here we can see explicitly how the blowup separates lines: it enlarges (or "blows up") first order differential data (the direction of a line through the origin) at p so much that it becomes topological data.

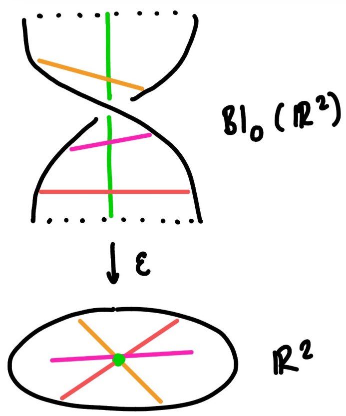

Figure 3.3: The blowup of $\mathbb{R}^2$ at a point (equivalently, the real points of $\mathbb{C}^2$ blown up at a point)

We now define the blowup of a projective surface more generally, defining the blowup of S at a point p as follows. Consider an open affine U containing p. Since it is smooth, it has a system of local parameters x, y (so the curves cut out by x and y are smooth at p and intersect transversely). Then the blowup $\widehat{U}$ of U at a point $p \in U$ is defined as the subvariety of $S \times \mathbb{P}^1$ (with $\mathbb{P}^1$ having coordinates $[X : Y]$) cut out by the equation

$$xY - yX = 0$$

Away from $\varepsilon^{-1}(p)$, ε is one-to-one, and in fact defines an isomorphism of $\widehat{U} - \varepsilon^{-1}(p)$ and $U - p$. The fiber of the point p is isomorphic to a $\mathbb{P}^1$, parametrizing the lines in U through p. Then the blowup at $\widehat{S}$ is defined by gluing $\widehat{U}$ and $S - p$ via the isomorphism $\varepsilon : \widehat{U} - \varepsilon^{-1}(p) \to U - p$. If S is smooth, then the blowup is smooth as well.

It turns out that the *only* obstruction to extending a rational map of smooth surfaces to a regular map is this lack of differential data being captured at points. We will soon show that given a rational map of smooth surfaces $\varphi : S \dashrightarrow S'$, we can construct a surface $\widehat{S}$ as a sequence of blowups of S such that the map $\widetilde{\varphi} : \widehat{S} \to S'$ is regular and agrees with φ away from the blown up points. In this way, the birational geometry of surfaces is still extremely rigid, and extremely fruitful to study in our goal to understand the classification of surfaces up to biregular morphism.

3.3 Properties of Blowups

Here, we state and prove a few properties of blowups. Let S be a smooth surface, and let $\varepsilon : \widehat{S} \to S$ be the blowup at a point p. We define the *exceptional divisor* E of this blowup to be the divisor corresponding to $f^*([p])$. Often times, we will be interested in the behavior of the pullback of curves under such a map. For a curve C intersecting p, define the *strict transform* $\widehat{C}$ of a curve C as the closure of $\varepsilon^{-1}(C - p)$ in $\widehat{S}$. If C has multiplicity m at p, then letting u, v be local parameters of S at p, the local function f defining C has an algebraic power series expansion $f_m(u, v) + f_{m+1}(u, v) + \ldots$ in u, v, with f_k a homogenous polynomial in u, v of degree k. By the definition of the blowup, around $(p, [1:0])$ we can find local coordinates given by $\widetilde{u} = u \circ \varepsilon$ and t a local coordinate of $\mathbb{P}^1$ at $[1:0]$. In these coordinates, $\widetilde{u} = 0$ is the defining equation of the exceptional divisor. We see that the pullback of f to $\widehat{S}$ is of the form

$$f \circ \varepsilon(\widetilde{u}, t) = f(\widetilde{u}, t\widetilde{u}) = \widetilde{u}^m (f_m(1,t) + \widetilde{u} f_{m+1}(1,t) + \ldots)$$

The same calculation can be done on a local chart around $(p, [0:1])$, giving us that for $\widehat{S}$ the blowup of S at p and an irreducible curve C passing through p with multiplicity m,

$$\varepsilon^*(C) = \widehat{C} + mE \tag{3.1}$$

This result will allow us to understand the intersection theory of a blowup from that of the original surface. Given any two divisors D_1 and D_2 passing through p, we can find linearly equivalent divisors $\widetilde{D_1}, \widetilde{D_2}$ (not necessarily effective) that avoid p. Since the blowup is an isomorphism away from p, we get that

$$\varepsilon^*(D_1).\varepsilon^*(D_2) = D_1.D_2$$

With this, we can also compute the self-intersection number of E. Consider a curve C passing through p with multiplicity 1, then it's strict transform $\widehat{C}$ has intersection multiplicity 1 with E. But since we know that $\varepsilon^*(C) = \widehat{C} + E$ and that $\varepsilon^*(C).E = 0$ by moving C away from p, we get that

$$E^2 = -1$$

In light of the interpretation of the intersection number as the degree of the normal bundle, E is algebraically "stuck" and we cannot find another effective divisor linearly equivalent to it (suppose there was such a D, which we can modify so that it shares no common components with E. Then since both E and D are effective they must have nonnegative intersection number). Homologically we can perturb E, but it would have to intersect E in the reverse orientation of the surface itself.

Topological interpretation of the blowup: While we have given an algebraic construction of the blowup of a smooth projective surface S, we can also define it topologically. The algebraic blowup of $\mathbb{C}^2$ at the origin can be shown to be diffeomorphic to $\mathbb{CP}^2$ minus the origin $[0:0:1]$, with a reversed orientation such that the divisor $[E]$ corresponding to the line at infinity $[t:u:0]$ has self-intersection -1. In the same way as the algebraic blowup separates points, travelling on different lines in the chart $[x:y:z] : z \neq 0$ limits to different points on the line at infinity. Then we can define the topological blowup of S at a point p to be $S \# \overline{\mathbb{CP}^2}$, wher the bar denotes reversed orientation. It is clear that $\widehat{S}$ is diffeomorphic to $S \# \overline{\mathbb{CP}^2}$.

Lemma 10. There is an isomorphism $\mathrm{Pic}(S) \oplus \mathbb{Z} \to \mathrm{Pic}(\widehat{S})$ given by sending (D, n) to $\varepsilon^* D + nE$.

Proof. This map is obviously surjective, since every divisor in $\widehat{S}$ not containing the exceptional divisor $\varepsilon^{-1}(p)$ is the linear combination of strict transforms of divisors from S. To show that this map is injective, note that if $\varepsilon^*(D) + nE = 0$, then taking intersection products with E forces n to be zero. If $\varepsilon^*(D)$ is linearly equivalent to zero, we now show that D is linearly equivalent to zero. We may assume that D does not pass through p. Now by definition of being linearly trivial, there is a rational function $f : \widehat{S} \dashrightarrow \mathbb{C}$ who's corresponding divisor is $\varepsilon^*(D)$. Now f defines a rational function $f : S - p \dashrightarrow \mathbb{C}$ that is holomorphic in a punctured analytic neighborhood of p, so f extends to a meromorphic function on all of p. The divisor of this function is exactly D, so D must be linearly equivalent to zero as well. □

Lemma 11. There is an isomorphism $NS(S) \oplus \mathbb{Z} \to NS(\widehat{S})$ sending $(PD([D]), n)$ to $\varepsilon^* PD([D]) + nPD([E])$

Proof. Firstly note that $\varepsilon^* PD([D]) = PD([\varepsilon^* D])$ since the c_1 map is functorial, so surjectivity follows analogously to the previous Lemma. By the fact that the topological intersection form agrees with the algebraic intersection form, if $PD([\varepsilon^*(D)]) + nPD(E) = 0$, we must have $n = 0$. Now the fact that $\varepsilon^* PD([(D)]) = 0$ implies that $PD([D]) = 0$ follows from perturbing D away from the point we blow up at and using the topological description of the blowup. □

Lemma 12. For $K_S, K_{\widehat{S}}$ denoting the canonical classes of S and $\widehat{S}$ respectively, we have that

$$K_{\widehat{S}} = \varepsilon^* K_S + E$$

Proof. Since ε is an isomorphism away from the fiber of the blown up point, we have that $K_{\widehat{S}} = \varepsilon^* K_S + mE$ for some m. The value of this m follows from the adjunction formula for E inside S: since E has genus zero we get that

$$-2 = E^2 + E.K_{\widehat{S}}$$

But $E.K_{\widehat{S}} = E.\varepsilon^* K_S - m$, and $E.\varepsilon^* K_S = 0$ since we can perturb K_S to not be supported at the point we blow up at. All in all, we have that $-2 = -1 - m$, and so $m = 1$. □

3.4 Rational Maps of Surfaces Factor Through Blowups

In this section, we give a sketch of the fact that every birational map factors through blowups, precisely stated in the following theorem:

Theorem 10. (Structure of birational maps of surfaces) Let $\phi : S \dashrightarrow S'$ be a birational map of surfaces. Then there is a surface $\widehat{S}$ and morphisms $f : \widehat{S} \to S$, $g : \widehat{S} \to S'$ a composition of blow-ups and isomorphisms so that the following diagram commutes:

$$\begin{tikzcd} & \hat{S} \arrow[dl, "f"'] \arrow[dr, "g"] & \\ S \arrow[rr, dashed, "\phi"'] & & S' \end{tikzcd}$$

As discussed before, a rational map of surfaces is defined everywhere except for a finite set of points. The first goal in proving Theorem 10 is to show that any of these holes can be "filled in" with a blowup to turn a rational map into a regular map.

Lemma 13. (Elimination of Indeterminacy) Let $\phi : S \dashrightarrow X$ be a rational map from a surface to a projective variety. Then there exists a surface S', a morphism $\nu : S' \to S$ a composition of blowups, and a morphism $f : S' \to X$ such that the obvious diagram commutes.

Let $P \subset |D|$ be a linear system on S. Throughout the proof of this lemma and later in the paper, we will refer to a curve C in S as a *fixed curve* of P if every divisor in P contains C as a common component. In this setting, the rational map $\varphi_P : S \dashrightarrow \mathbb{P}^n$ induced by P will not be defined along C, and the image of φ will lie in a hyperplane of $\mathbb{P}^n$ (by the coordinate-free description of this map in Section 3.1). If we subtract enough factors of C from every divisor in P so that C is no longer a fixed curve of the corresponding linear system $P' \subset |D - mC|$, then $\varphi_{P'}$ (the induced map to projective space given by P') extends φ_P. If a linear system on a surface has no fixed curve, it's set of basepoints must be a finite set of points.

Proof of Lemma 13. We can assume that $X = \mathbb{P}^m$ and that $\phi(S)$ does not lie in a hyperplane of $\mathbb{P}^m$. Then the map $\phi : S \dashrightarrow X$ corresponds to a linear system $P \subset |D|$ with no fixed curves. If this linear system has no basepoints, then ϕ is a morphism and we are done. Otherwise, the number of basepoints of P is finite and bounded by D^2 since basepoints must lie in the intersection of any two divisors in P.

We blow up at S one of these basepoints of P, giving us a blowup morphism $\varepsilon_1 : S_1 \to S$. The rational map $\phi \circ \varepsilon_1 : S_1 \dashrightarrow X$ is induced by the linear system $\varepsilon_1^* P \subset |\varepsilon^*(D)|$ by (3.1), and has E as a fixed curve as p was a basepoint of P. After subtracting kE from every divisor in P, the linear system $P_1 = \varepsilon_1^* P - kE \subset |\varepsilon^*(D) - kE|$ has no fixed curves and induces a rational map that agrees with $\phi \circ \varepsilon_1$ on it's domain of definition.

Since P_1 has no fixed curves it has a finite number of basepoints, and this number is bounded by $(\varepsilon^*(D) - kE)^2$. The key observation is that

$$(\varepsilon^*(D) - kE)^2 = D^2 - k^2 < D^2$$

so the number of basepoints ϕ_{P_1} can possibly have is strictly less than the number of basepoints ϕ_P can have. Therefore we may repeat this process of the previous two paragraphs, blowing up basepoints until we have constructed a linear system P_m on a surface S_m such that the induced map $\varphi_{P_m} : S \dashrightarrow \mathbb{P}^n$ is an honest morphism and $\varphi_{P_m} = \phi \circ \varepsilon_1 \circ \cdots \circ \varepsilon_m$. $\square$

Lemma 13 allows us to obtain that for a birational map $S \to S'$ of surfaces, then there exists surface S'', a morphism $g : S'' \to S'$, and a commutative diagram

$$\begin{tikzcd}
S'' \arrow[d, "f"'] \arrow[dr, "g"] & \\
S \arrow[r, dashed, "\phi"'] & S'
\end{tikzcd}$$

where f is a composition of blowups. Thus, we see that g is a *birational* morphism from S'' to S'. To prove Theorem 10, it remains to understand the structure of g, which will be completed in the following lemma:

Lemma 14. Let $f : S \to S'$ be a birational morphism of surfaces. Then there exists a surface $\widehat{S'}$ constructed as a sequence of blowups of S', such that S is isomorphic to $\widehat{S'}$ and the obvious diagram commutes.

In the proof of Lemma 14, we will use the universal property of blowups. The proof is not complicated, but takes us away from our current discussion and so we refer the reader to II.8 of [Bea96].

Theorem 11. (Universal Property of Blowups) Any birational morphism $f : S \to S'$ of surfaces where f^{-1} is undefined at a point p factors as

$$S \xrightarrow{g} \widehat{S'} \xrightarrow{\varepsilon} S'$$

where g is a birational morphism, $\widehat{S'}$ is the blowup of S' at p, and ε is the corresponding blowup map.

Proof of Lemma 14. If f is an isomorphism we are done. If not, there is a point p such that f^{-1} is not defined at p. Let $\varepsilon_1 : S'_1 \to S'$ be the blowup of S' at p. Then by the universal property of blowing up, $f : S \to S'$ factors as $\varepsilon_1 \circ f_1$ where f_1 is a birational morphism. If f_1 is not invertible, we may repeat this process, constructing blowups $\varepsilon_i : S'_i \to S'_{i-1}$ and birational morphisms f_i such that $f_{i-1} = \varepsilon_i \circ f_i$ until f_i is an isomorphism

We now show that this process must terminate. First note that f contracts a finite number of curves in S, as any such curve must be in the fiber of a point not in the domain of definition of f^{-1}. Since ε_1 contracts the exceptional divisor E_1 of the corresponding blowup, f_1 contracts strictly less curves than f does. By the same reasoning, f_{i+1} contracts strictly less curves than f_i, and eventually, this number will be zero and f_i will be an isomorphism for large enough i. □

Combining Lemmas 13 and 14 immediately proves Theorem 10 on the structure of birational maps of surfaces.

With the structure of birational maps of surfaces, we can now connect any two surfaces S_1, S_2 in the same birational equivalence class to a third surface S_3 which they both blow up to. With this, we can define the notion of a minimal surface - in a sense a "simplest" representative of every birational equivalence class.

Definition 9. We define a *minimal surface* as a smooth surface that isn't obtained as a blowup of another surface.

Importantly, we have the following result:

Lemma 15. Every surface can be attained as a finite sequence of blowups of a minimal surface

Proof. Consider a surface S. If it is not minimal, there exists a surface S' where S is isomorphic to a blowup of S' at a point. If S' is not minimal, there exists a surface S'' where S is isomorphic to to the blowup of S'' at a point. We continue this process until the surface we attain is minimal. This process must terminate as every time we blow up a surface, the rank of the Néron-Severi group increases by

1. Since the Néron-Severi group of our starting S is finitely generated, we can blow down S at most $\operatorname{rk}(NS(S))$ many times before reaching a minimal surface. □

As we've seen, blowups are extremely well-behaved in how they change the algebraic and geometric data attached to a surface. The previous theorem allows us to reduce the classification of surfaces to the classification of minimal surfaces, as we can understand all surfaces as blowups of these minimal surfaces.

Finally, we make a quick observation about when blowups are in fact unnecessary:

Lemma 16. Every morphism f from $\widehat{S}$ to a projective variety X that contracts the exceptional divisor E to a point must also factor through S.

Proof. We can assume $X = \mathbb{P}^n$, then pick an affine chart of $\mathbb{P}^n$ and shrink the domain so that we have a map $U \longrightarrow \mathbb{A}^n$, where $U \ni p$ is an open set of S. By looking at individual coordinate functions we can reduce the problem to showing that any map $U - p \longrightarrow \mathbb{A}^1$ extends across U. But this is just Hartog's lemma, which holds in both the complex and algebraic setting. □

3.5 Castelnuovo's Contractibility Criterion

Now that we know that every surface dominates a minimal surface, we seek a practical way to determine whether or not a surface is minimal, and if not, how to turn it into one.

We know that the exceptional divisor of a blowup is a smooth rational curve of self-intersection -1. It turns out that any time we have a surface S and find such a smooth rational curve E with $E^2 = -1$, S must be the blowup of some smooth surface. Precisely, there must be another surface S' and a point $p \in S'$ such that given the blowup $\varepsilon : \widehat{S'} \longrightarrow S'$ at p, $\widehat{S'} \cong S$ and under this isomorphism the exceptional divisor of ε corresponds to $E \subset S$. This fact is known as Castelnuovo's Contractibility Criterion.

Theorem 12. (Castelnuovo's Contractibility Criterion) Let S be a surface and $E \subset S$ a smooth rational curve with $E^2 = -1$. Then S is the blowup of another surface S' and E is the exceptional divisor of this blowup.

This theorem gives a characterization of minimal surface as exactly those that don't contain smooth rational curves of self-intersection -1. Furthermore, this theorem gives us a procedure on how to construct minimal surfaces: given a surface S, we search for a -1 rational curve. If we cannot find such a curve, S must be minimal. If we can find one, the above theorem will produce for us an S' that blows up to S, and we repeat the process with S'.

The proof of Theorem 12 has two parts. The first part of the proof involves explicitly constructing a morphism φ from S to some projective space, such that φ is an isomorphism on $S - E$, but that E is contracted to a point. The second part of the proof involves checking that $\varphi(S)$ is a smooth surface, and is a fairly technical topological argument. We will give an exposition of the construction of the morphism φ, and leave the details of checking that $\varphi(S)$ is smooth to the reader (all of which can be found in II.17 of [Bea96]).

Proof of the construction of the contraction. The main idea of the construction of φ comes twisting one line bundle by a second sufficiently positive line bundle, which has a very geometric meaning. On the structure sheaf of any compact complex manifold in $\mathbb{P}^n$, the only global sections are the constant functions, so working directly with the structure sheaf in any cohomological arguments shouldn't be very effective. By considering the pullback of $O(1)$ of $\mathbb{P}^n$ onto S, $O_S(1)$ encodes "hyperplane sections of S". By increasing the degree of this line bundle, say from $O_S(1)$ to $O_S(2)$, the sections of this line bundle are now encoding "quadric sections of S" (or equivalently, considering hyperplane sections of S embedded into some larger $\mathbb{P}^N$ via the second Veronese embedding). We should think about twisting a line bundle by a positive degree divisor as giving ourselves more sections to capture the geometry of a variety more finely. This is the rough picture for why Serre vanishing is true, which we will use in our proof.

With this, given any hyperplane section H_0 of S, by Serre vanishing we can find a large enough n such that $H^1(S, O_S(nH_0)) = 0$. For notational convenience, let $H = nH_0$. Now for D an effective divisor, if the map to projective space associated to $O_S(D)$ were to contract the curve E, we would need that $D.E = 0$ (so that the hyperplane of divisors in $|D|$ passing through a point $p \in E$ would be the same no matter which p we choose). Therefore, setting $k = H.E$, we modify H to get a divisor $H' = H + kE$ for which $H'.E = 0$. The rest of the proof will be showing that the map to projective space associated to $O_S(H')$ gives the desired contraction φ we seek. The fact that nH is very positive gives us hope that even after modifying by kE, there are enough sections of $O_S(H')$ such that the associated map still is embedding of S away from E.

We consider the residue exact sequence

$$0 \to O_S(-E) \to O_S \to O_E \to 0$$

The last entry of the exact sequence above is O_E since $E = \mathbb{P}^1$, so as $E^2 = 0$, $O_E(E)$ has degree 0 and thus is just O_E. Twisting the above sequence by $H + iE$ for $1 \leq i \leq k$ gives us the short exact sequence

$$0 \to O_S(H + (i-1)E) \to O_S(H + iE) \to O_E(k - i) \to 0$$

and we can consider the corresponding long exact sequence

$$\begin{aligned} 0 \to H^0(S, O_S(H + (i-1)E)) \to H^0(S, O_S(H + iE)) \to H^0(E, O_E(k-i)) \\ \to H^1(S, O_S(H + (i-1)E)) \to H^1(S, O_S(H + iE)) \to 0 \end{aligned}$$

(where the last 0 is from Serre Duality). For $i = 1$, since $H^1(S, O_S(H)) = 0$ by assumption, $H^1(S, O_S(H+E)) = 0$ as well, and we may induct to conclude that $H^1(S, O_S(H + kE)) = 0$.

Now for all i, we have the exact sequence

$$0 \to H^0(S, O_S(H + (i-1)E)) \to H^0(S, O_S(H + iE)) \to H^0(E, O_E(k-i)) \to 0 \tag{3.2}$$

We build up a basis of the vector space $H^0(S, O_S(H+kE))$ inductively, starting with a basis of $s_0, \ldots, s_n$ of $H^0(S, O_S(H))$. Let t be a section of $O_S(E)$ who's divisor of zeros is E. Setting $i = 1$ in (3.2) we can build a basis of $H^0(S, O_S(H + E))$ as $ts_0, \ldots, ts_n$ and a lift of a basis $H^0(O_E(k-1))$. Inductively, we get a basis of $H^0(S, O_S(H + kE))$ as

$$\left\{t^k s_0, \ldots, t^k s_n\right\} \cup \left\{t^{k-1} \times \text{lift of basis of } H^0(E, O_E(k-1))\right\} \cup \ldots$$
$$\cup \left\{t \times \text{lift of basis of } H^0(E, O_E(1))\right\} \cup e$$

where e is the lift of the single generating element of $H^0(E, O_E)$. Now these sections give us a rational map from S to some large $\mathbb{P}^N$. Since the basis of $H^0(S, O_S(H))$ is basepoint-free and t only vanishes on E, the basis above is basepoint-free away from E. In fact, it is basepoint-free on E as well as e is constant and nonzero on E, so the basis above gives a well-defined morphism of S into $\mathbb{P}^n$. Since the basis $s_0, \ldots, s_n$ gives an embedding of S to $\mathbb{P}^n$, we see that our map into $\mathbb{P}^N$ defines an embedding away from E, and it is clear from the definition of the morphism that E is contracted to $[0 : \ldots, 0 : 1]$.

Once we show that $\varphi(S) = S'$ is smooth, then Lemma 14 tells us that S is isomorphic to a sequence of blowups on S'. The rank of the Neron-Severi group of S' is at most one less than S as we only contracted the class of a single curve, but it cannot be equal since Lemma 14 would then tell us that $S \cong S'$. Therefore, by Lemma 11 we see that S is isomorphic to S' blown up once. □

The operation of taking S and contracting a smooth rational -1 curve to obtain a simpler smooth surface S' is called a *blowdown*, and is the inverse of the blowup.

Interestingly, there are examples of surfaces that contain infinitely many rational curves of self-intersection -1, but thankfully by Lemma 15 we only need to blow down a finite number of times to be able to reach a minimal surface. We can construct one as follows: take two smooth degree three curves D_1, D_2 in $\mathbb{P}^2$ (i.e. tori by the degree-genus formula) intersecting at 9 distinct points. The pencil defined by these two divisors gives a rational map from $\mathbb{P}^2$ to $\mathbb{P}^1$ with basepoints at the 9 points of intersection. Blowing up at these basepoints gives us a surface $\widehat{S}$ and a morphism $\varphi : \widehat{S} \to \mathbb{P}^1$. See Figure 3.4 for an illustration. The 9 exceptional divisors corresponding to the blowups give sections of φ, since they intersect every divisor in the pencil exactly once. Every fiber of φ is a smooth complex torus, thus an elliptic curve. Choose a marked point in each torus fiber by their intersection with a fixed exceptional divisor E_1 (in light blue in Figure 3.4), giving each fiber a group structure. Now we fix a different exceptional divisor E_2 (in purple in Figure 3.4), intersecting each fiber F_i at a point p_i. Translating each fiber F_i by the element p_i gives an automorphism of $\widehat{S}$. It can be shown that since $\mathrm{Pic}(\widehat{S})$ is torsion free, the automorphism described above for general enough cubics is non-torsion in $\mathrm{Aut}(\widehat{S})$. Then the orbit of any of the remaining -1 sections gives infinitely many rational curves of self-intersection -1 in $\widehat{S}$.

With this, we conclude our discussion of blowups and begin to investigate families of surfaces.

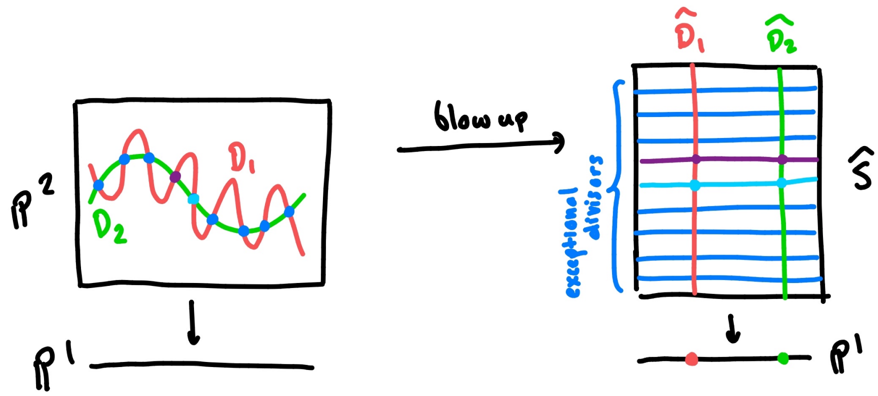

Figure 3.4: The construction of a surface with infinitely many rational curves of self-intersection -1

Chapter 4

Ruled and Rational Surfaces

We start our study of surfaces by considering some of the simplest ones: the ruled and rational surfaces. A *ruled surface* is a surface birational to $C \times \mathbb{P}^1$ for C a smooth curve. A *rational surface* is a surface birational to $\mathbb{P}^2$. For both these families, we seek to understand their minimal models as well as their intersection theory. As $\mathbb{P}^1 \times \mathbb{P}^1$ is birational to $\mathbb{P}^2$, the rational surfaces are a subclass of the ruled surfaces. We distinguish the rational surfaces as we will have to handle the classification of their minimal models differently than the "irrational" ruled surfaces (those birational to $C \times \mathbb{P}^1$ where $C \neq \mathbb{P}^1$).

First, we gather a general fact about the cohomology of ruled surfaces that we will use repeatedly:

Lemma 17. For a ruled surface S birational to $S \times \mathbb{P}^1$, $h^1(S, \mathcal{O}_S) = g(C)$.

Proof. Firstly, we show that $h^1(S, \mathcal{O}_S)$ is a birational invariant. Note that by the Dolbeault theorem, we have that $h^1(S, \mathcal{O}_S) = h^0(S, \Omega^1(S))$, and so we show that $h^0(S, \Omega^1(S))$ is a birational invariant. Consider a rational map $S \dashrightarrow S'$, i.e. a morphism $\varphi : S - F \to S'$ where F is a finite set of points. Take a holomorphic (equivalently, regular) form ω on S', then the pullback $\varphi^*\omega$ will be a differential form on S with potential poles at F. However, the poles of $\varphi^*\omega$ are divisors on S, i.e. codimension 1, and thus $\varphi^*\omega$ cannot have any poles and must be regular on all of S. Doing the same in the other direction shows that $H^0(S, \Omega^1_S) \cong H^0(S', \Omega^1_{S'})$.

From here, it is sufficient to compute $h^1(S, \mathcal{O}_S)$ for $C \times \mathbb{P}^1$. By the Kunneth formula, we see that $h^1(S, \mathbb{R})$ is $2g(C)$. Now the Hodge decomposition on first cohomology gives us that

$$h^1(S, \mathcal{O}_S) = h^{0,1}(S) = \frac{1}{2} h^1(S, \mathbb{R})$$

We conclude $h^1(S, \mathcal{O}_S)$ is $g(C)$ as desired. □

We generate some examples of ruled surfaces. Aside from the obvious $C \times \mathbb{P}^1$, we also can consider $\mathbb{P}^1$ bundles, i.e projectivizations of rank 2 vector bundles over C. Since Zariski opens are dense in C, a trivializing neighborhood of a $\mathbb{P}^1$ bundle over C gives a birational map to $C \times \mathbb{P}^1$.

A particular class of surfaces that will be important to our studies are the *geometrically ruled surfaces*, those that admit a smooth map $S \to C$ with rational fibers. Geometrically ruled surfaces are especially nice to work with, and one such reason for this is the following:

Lemma 18. Every homology class of a geometrically ruled surface is algebraic, i.e. the map from $\mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$ from the exponential exact sequence is surjective.

Proof. To show this, it's sufficient to show that $H^2(S, \mathcal{O}_S) = 0$. Take a fiber F over a point p of the map $\pi : S \to C$. We have that $F^2 = 0$ by our usual reasoning: take a divisor D' on C linearly equivalent to p but disjoint from it, then $F^2 = \pi^*(p).\pi^*(D') = 0$. It follows by adjunction and the fact that F is rational that $F.K = 0$. If $H^2(S, \mathcal{O}_S)$ was nontrivial, by Serre duality $|K|$ would be nonempty, so there is an effective divisor D with $D.F = 0$. But breaking up D into $nF + D'$ with D' effective and not containing F as a component, we see that this cannot be possible (due to the fact that F^2 is nonnegative). □

A concrete example of geometrically ruled surfaces is a projectivization of rank 2 vector bundles over a curve. This example was also ruled as discussed earlier, so it is a natural question to ask if all geometrically ruled surfaces are in fact ruled. The answer will turn out to be yes, and in fact all geometrically ruled surfaces will be projectivizations of rank 2 vector bundles over some curve C. To show this requires us to understand the structure of $\mathbb{P}^1$ fibrations of curves, which is the content of the Noether-Enriques Theorem.

4.1 The Noether-Enriques Theorem

A structural tool that we will need in the study of ruled surfaces is the Noether-Enriques theorem. The question we seek to answer is:

> Given a surjective map $S \to C$ smooth over a point $x \in C$ who's fiber is $\mathbb{P}^1$, what can we say about nearby fibers and how they vary?

Things behave as nicely as possible if we view S and C as smooth manifolds rather than complex ones: we can find a neighborhood of x where every fiber is $\mathbb{P}^1$ and locally the map is just projection:

Lemma 19. Let $f : M \to N$ be a proper surjection, and let x be a regular value in N (i.e. a point where f is a submersion on it's fiber). Then there is a neighborhood U of x in N such that $f^{-1}(U)$ is a trivial fibration over U (in the smooth category) with fibers diffeomorphic to $f^{-1}(U)$.

Proof. The set of points in M where f is surjective on tangent spaces is open in M, thus by properness and surjectivity the set of regular points of f in N is open as well. Now restrict f to the preimage of the regular values in N, and use Ehresmann's lemma in differential topology (Lemma 2). □

But what can we say when we're working over the complex analytic or algebraic categories? In particular, in the algebraic category, our open sets in the Zariski topology are very coarse, so even if we are able to show that a neighborhood of x has $\mathbb{P}^1$ fibers, it could be the case that we can't pick a "small enough" neighborhood for the fibration to be locally trivial. The Noether-Enriques theorem tells that for surfaces, we can always locally trivialize:

Theorem 13. (Noether-Enriques) Let S be a surface, π a surjective morphism from S to a curve C, $x \in C$ a point where π is smooth and the fiber $\pi^{-1}(x)$ is isomorphic to $\mathbb{P}^1$. Then there exists a Zariski open U of C containing x where $\pi^{-1}(U) \cong U \times \mathbb{P}^1$ and this isomorphism commutes with projection.

Not that this is *not true* in general for varieties of higher dimension, exactly for the reason that $\mathbb{P}^1$ fibrations do not have to be locally trivial in the Zariski topology. For example, every conic in $\mathbb{P}^2$ is of the form $Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz$, so we can consider the *universal conic* parametrizing all smoooth conics. We construct this by considering the variety X of $\mathbb{P}^5 \times \mathbb{P}^2$ given by points $([A : B : C : D : E], [x_0 : y_0 : z_0])$ where $Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz$ is a smooth conic and the conic evaluates to zero on $[x_0 : y_0 : z_0]$. Now all smooth conics in $\mathbb{P}^2$ are isomorphic to $\mathbb{P}^1$ (the projection map from a point on the conic to a hyperplane is birational and thus an isomorphism), so once we delete the fibers of points in Δ, the set $[A : B : C : D : E]$ of $\mathbb{P}^5$ where the corresponding conic is singular, we get another variety X' that is a $\mathbb{P}^1$ fibration over $\mathbb{P}^5 - \Delta$. Note that X is a smooth surface, which can easily be checked by computing the partials of $Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz$ with respect to the 9 variables, and therefore X' is smooth too. Since a morphism of smooth varieties with fibers all the same dimension is flat, we have that all fibers are numerically equivalent to each other.

We now show that X' cannot be a $\mathbb{P}^1$ bundle over $\mathbb{P}^5 - \Delta$. To see this, suppose for contradiction that there existed a trivializing Zariski-open neighborhood U of $\mathbb{P}^5 - \Delta$, then there would be a regular section s such that $V = s(U)$ intersects every fiber of U exactly once. Now we take the closure of V in X. The fiber of the point representing the conic $x^2 = 0$ is a $\mathbb{P}^1$ with multiplicity 2, and therefore $\overline{V}$ must intersect this particular fiber an even number of times - see Figure 4.1 for an illustration. But all fibers of $\pi : X \to \mathbb{P}^5$ are numerically equivalent, so every fiber of $U \in \mathbb{P}^5 - \Delta$ has to intersect V with even multiplicity, a contradiction.

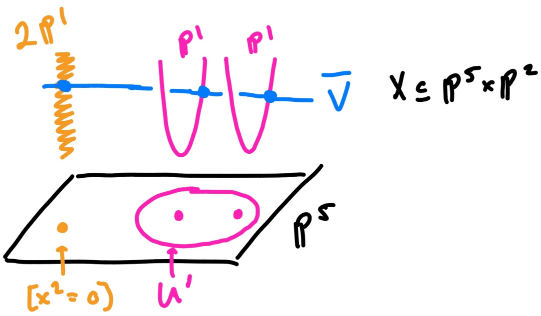

Figure 4.1: An illustration of a counterexample of Noether-Enriques for higher-dimensional varieties

The main strategy of proof for the Noether-Enriques theorem will be to find a divisor of S that intersects the fiber of $x \in C$ exactly once. We will use this "local section" to then trivialize the fibration in a neighborhood of X.

Proof of Noether-Enriques. From Lemma 18, the exponential exact sequence gives a surjective map from $\mathrm{Pic}(S) = H^1(S, \mathcal{O}_S^*) \to H^2(S, \mathbb{Z})$, and we search for a cohomology class $h \in H^2(S, \mathbb{Z})$ intersecting $f = c_1(\mathcal{O}_S(F))$ once (i.e. $h.f = 1$). The image of the map $H^2(S, \mathbb{Z}) \to \mathbb{Z}$ given by dotting with f has image an ideal $d\mathbb{Z}$ of $\mathbb{Z}$, and we need to show that $d = 1$. Consider the linear map $H^2(S, \mathbb{Z}) \to \mathbb{Z}$ sending a to $\frac{1}{d}(a.f)$. Since Poincaré duality is a perfect pairing, this tells us that

$$H^2(S, \mathbb{Z}) \to \mathrm{Hom}(H^2(S, \mathbb{Z}), \mathbb{Z}) \qquad b \to (f \mapsto (b.f))$$

is surjective, with kernel the torsion subgroup of $H^2(S,\mathbb{Z})$. This tells us that up to torsion, there exists some $f' \in H^2(S,\mathbb{Z})$ with $f = df'$. If $d > 1$, this seems off, since it would mean $F = dF'$ for some divisor F', and F should have multiplicity 1 by smoothness. We can show a contradiction by considering the image $k \in H^2(S,\mathbb{Z})$ of the canonical bundle, and noting that $b^2 + b.k$ is always an integer and even (using the adjunction formula). Plugging in f', using that $f = df'$, and applying adjunction on f gives us that

$$f'^2 + f'.k = -\frac{2}{d}$$

is an even integer, and so $d = 1$.

We now have a divisor H such that $H.F = 1$, and this will be our "local trivializing section" for our trivializing neighborhood. We hope to build a linear system that "stratifies" this H, giving us an open set $U \subset C$ an isomorphism of $\pi^{-1}(U)$ with $U \times \mathbb{P}^1$. See Figure 4.2 for an illustration. However, H doesn't need to be effective, so we need to modify H to achieve this. One obstruction for a line bundle $\mathcal{O}_S(D)$ to have sections is for $\mathcal{O}_C(D)$ to have nonnegative degree - which we can guarantee by boosting the degree of $D|_C$ on C. Since F has degree 1 on C, we add copies of F to H to try to make $h^0(H + kF)$ positive for large enough k.

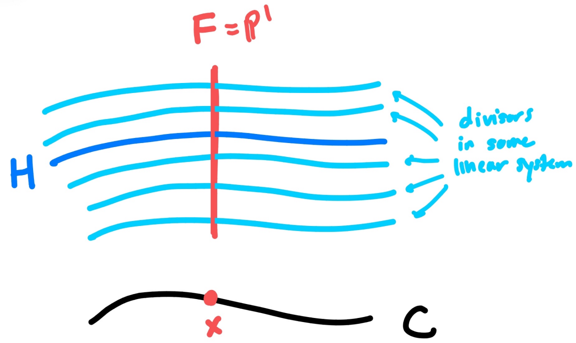

Figure 4.2: "Stratifying" our H to get a trivialization

In this vein, we twist the residue exact sequence $\mathcal{O}_S \to \mathcal{O}_S(F) \to \mathcal{O}_F \to 0$ by $H + (r-1)F$ and take long exact sequences to get

$$H^0(S, \mathcal{O}_S(H+rF)) \to H^0(F, \mathcal{O}_F(1)) \to H^1(S, \mathcal{O}_S(H+(r-1)F)) \to H^1(S, \mathcal{O}_S(H+rF)) \to 0$$

For large enough r, $H^1(S, \mathcal{O}_S(H+(r-1)F)) = H^1(S, \mathcal{O}_S(H+rF))$ since the map $H^1(S, \mathcal{O}_S(H+(r-1)F)) \to H^1(S, \mathcal{O}_S(H+rF))$ is always surjective. Therefore, for large enough r the map $H^0(S, \mathcal{O}_S(H+rF)) \to H^0(F, \mathcal{O}_F(1))$ is surjective. We can pick two linearly independent sections of $\mathcal{O}_S(H+rF)$ that surject onto $\mathcal{O}_F(1)$, giving us a linear pencil P spanned by these two section who's corresponding divisors will form the stratifications in Figure 4.2. Since $\mathcal{O}_F(1)$ is already basepoint free, the restriction of the pencil P to F is basepoint free as well. Since the image of any curve in S is either a point or all of C, any fixed curve in P must necessarily lie in a fiber of π. The basepoints of P away from fixed curves are finite and lie in a set of fibers distinct from F, so all in all, once we account for all points $x_i \in C$ who's fibers contain fixed curves or basepoints of P, the open set $U = C - \cup_i x_i$ is such that P is basepoint free on $\pi^{-1}(U)$. Since π is smooth over x, we can shrink U to a smaller Zariski open set such that every point in U is a regular value of π. Since the map $\pi^{-1}(U) \to U$ is a smooth submersion, all of it's fibers are homologous.

Since any divisor C' in the restriction of $P|_{\pi^{-1}(U)}$ intersects F once and all fibres are homologous, we see that C' must be a formal sum of a section of π and a potentially nonempty collection of fibers. But if C' contained a fiber, then any other divisor in $P|_{\pi^{-1}(U)}$ would have to intersect C' positively, creating basepoints. We conclude that all divisors in $P|_{\pi^{-1}(U)}$ are sections of π over U, and we have a map $\phi : P|_{\pi^{-1}(U)} \to \mathbb{P}^1$.

Now we can check that the map $(\pi, \phi) : p^{-1}(U) \to U \times \mathbb{P}^1$ is an isomorphism and that the diagram

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr, "(\pi,\phi)"] \arrow[dr, "\pi"'] & & U \times \mathbb{P}^1 \arrow[dl, "\text{proj}"] \\
& U &
\end{tikzcd}$$

commutes, completing the proof.

□

From Noether-Enriques it immediately follows that every geometrically ruled surface is ruled.

> **Remark .** In fact, for $\varphi : S \to C$ a morphism with S a smooth surface and C a smooth curve, all fibers of $S \to C$ are homologically equivalent, thus numerically equivalent! To see this, first note that the set of regular points of $\varphi : S \to C$ is a Zariski open set U of C, and we already know that over U all fibers are homologically equivalent. Let $p_1, \ldots, p_n$ be the points in the complement of U. Pick some p_i, then we can find a rational function on C with a simple pole at p_i and all other poles/zeros away from p_j for $j \neq i$[a]. Then pulling back through φ, we get that
>
> $$\varphi^{-1}([p_i]) = \sum_i n_i F_i$$
>
> for F_i fibers over U. Since $\sum_i n_i = 1$ and all the F_i are already homologous to each other, taking the c_1 map of both sides gives us that $\varphi^{-1}([p_i])$ is homologous, thus numerically equivalent, to any fiber over U.
>
> In the statement of Noether-Enriques (Theorem 13), the condition of our morphism π being smooth over the point x was only used in showing that the fibers of π are homologous. Thus in the setting where S, C are both smooth, we can omit the smoothness assumption on Noether-Enriques.
>
> [a] To find such a function, one way to do it is to embed C into $\mathbb{P}^n$, then pick two hyperplanes, one intersecting p_i once and avoiding the p_js, and one avoiding all of the complement of U.

If every fiber of our map is a $\mathbb{P}^1$, then by doing Noether-Enriques to get trivializing patches we the following easy corollary:

Corollary 4. Every $\mathbb{P}^1$ fibration over a curve C is the projectivization of a rank 2 vector bundle over C

Proof. A $\mathbb{P}^1$ fibration over a curve C is locally trivial by Noether Enriques. This means that we have commutative diagrams

$$\begin{tikzcd} \pi^{-1}(U_i) \arrow[r, "f_i", "\sim"'] \arrow[d, "\pi"'] & U_i \times \mathbb{P}^1 \arrow[dl, "p_i"] \\ U_i & \end{tikzcd}$$

where p_i is the projection map onto the first coordinate, and f_i is an isomorphism. Therefore, the transition functions between different local trivializations $U_i \cap \mathbb{P}^1$ and $U_j \cap \mathbb{P}^1$ are functions $U_i \cap U_j \to PGL_2(\mathbb{C})$, the automorphism group of $\mathbb{P}^1$. In this way, $\mathbb{P}^1$-fibrations over C can be identified with $H^1(C, PGL_2(\mathcal{O}_C))$. From the exact sequence

$$1 \to \mathbb{C}^* \to GL_2(\mathbb{C}) \to PGL_2(\mathbb{C}) \to 1$$

we have an exact sequence of sheaves

$$1 \to \mathcal{O}_C^* \to GL_2(\mathcal{O}_C) \to PGL_2(\mathcal{O}_C) \to 1$$

We take the long exact sequence, and noting that $H^2(C, \mathcal{O}_C^*) = 0$ (since $\dim C = 1$) gives us a surjective map $H^1(C, GL_2(\mathcal{O}_C)) \to H^1(C, PGL_2(\mathcal{O}_C))$. Note that this map is given by modding the transition functions of a rank 2 vector bundle by scalars, which is exactly the act of taking the projectivization of a rank 2 vector bundle. □

Note that the map $H^1(C, \mathcal{O}_C^*) \to H^1(C, GL_2(\mathcal{O}_C))$ in the long exact sequence considered sends a line bundle L to the trivial rank 2 vector bundle twisted by L. Therefore, the projectivization of a rank 2 vector bundle is invariant under twisting by a line bundle.

4.2 All Irrational Minimal Ruled Surfaces are Geometrically Ruled

Now that we have the general machinery of Noether-Enriques, we can attack the problem of classifying the minimal models of ruled surfaces. The picture will look slightly different for irrational ruled surfaces (i.e. those birational to $C \times \mathbb{P}^1$ for C irrational) and for the rational ruled surfaces, due to the simple fact that any map of projective curves from lower genus to higher genus must be constant (due to Riemann Hurwitz).

> **Theorem 14.** Let C be a smooth, irrational curve. Then the minimal models of $C \times \mathbb{P}^1$ are precisely the geometrically ruled surfaces over C.

Proof. Firstly, we show that geometrically ruled surfaces S with a projection map $\pi : S \to C$ are in fact minimal. If an exceptional divisor E lied entirely in a fiber of π, then we arrive at a contradiction: $E^2 = -1$ by definition, but as E is a fiber of π we must have that $E^2 = 0$ (again, we can find a divisor linearly equivalent to $\pi(E)$ disjoint from $\pi(E)$ and take pullbacks). Since the image of E is closed and connected in C, if E doesn't lie in a fiber we must have $\pi(E) = C$. However, since we assumed C to have higher genus than 0, by Riemann-Hurwitz the only maps from E to C are constant. We conclude that there cannot exist any exceptional divisors in S.

Now we show that every minimal surface of $C \times \mathbb{P}^1$ is geometrically ruled. Take such a minimal surface S, birational to $C \times \mathbb{P}^1$ under a map φ, and compose this map with the projection π from $C \times \mathbb{P}^1$ to C. By elimination of indeterminacy, the rational map $\pi \circ \varphi$ lifts to a sequence of blowup of S:

$$\begin{tikzcd} & \hat{S} \arrow[dl, "\varepsilon_n \circ \cdots \circ \varepsilon_1"'] \arrow[dr, "f"] & \\ S \arrow[rr, dashed, "\pi \circ \varphi"] & & C \end{tikzcd}$$

The exceptional divisor E_n of ε_n is contracted to a point through the regular map f, since E_n is a rational curve while C is irrational. Thus by Lemma 16, the last blowup ε_n is unnecessary for f to be regular. By induction, we actually get that $\pi \circ \varphi$ is regular. Now since a birational map of surfaces is an isomorphism away from a finite set of points, the fiber of $\pi \circ \varphi$ is generically $\mathbb{P}^1$.

All that's left to show is that not only is a generic fiber of $\pi \circ \varphi$ a rational curve, but in fact every fiber is. Choose any fiber F of $\pi \circ \varphi$. We have that $F^2 = 0$ and $F.K = -2$ (fibers isomorphic to $\mathbb{P}^1$ have this fact by the genus formula, now use that all fibers are numerically equivalent so this fact is true for all F). Now either

- F is of the form nD for some $n \in \mathbb{N}$ and D a curve. However, we must have that $n = 1$, or else in the proof of Noether-Enriques there would be no class intersecting F exactly once. The fact that $D^2 = 0, D.K = -2$, and the remark after the general genus formula (Corollary 3) gives us that F is in fact a smooth $\mathbb{P}^1$.

- F is of the form $\sum_{i=1}^k n_i D_i$ for $k > 1$ and $n_i \geq 0$. We first show that F is connected. To do this, we apply Stein Factorization on $\pi \circ \varphi$ to get that $\pi \circ \varphi$ breaks up as

  $$S \xrightarrow{\phi} C' \xrightarrow{\nu} C$$

  where ϕ has connected fibers and ν is a finite morphism. Since $\pi \circ \varphi$ has connected fibers on a Zariski open set of C, the degree of the finite map ν is 1 on a Zariski open set of C and thus is 1 everywhere. Thus, we conclude that $\pi \circ \varphi$ has connected fibers.

  As F is connected and reducible of the form $\sum_{i=1}^k n_i D_i$ for $k > 1$, we can show that $D_i^2 < 0$ for all i. To do this, note that we have that $n_i D_i^2 = D_i.(F - \sum_{j \neq i} n_j D_j)$, but $D_i.F = 0$ by perturbing the point the fiber lies over, and $D_i.D_j > 0$ for some $i \neq j$ since the map $\pi \circ \varphi$ has connected fibers.

  Using this fact along with the the general genus formula we get that

  $$(D_i.K) \geq 2p_a(D_i) - 1 \geq -1$$

  where p_a is the arithmetic genus of D_i. Therefore, $(D_i.K) \geq -1$, with equality if and only if $D_i^2 = -1$ and $p_a(D_i) = 0$. However, this would imply that D_i is a smooth rational curve of self-intersection -1 lying in S, a contradiction to S being minimal. Therefore, it must be the case that $D_i.K \geq 0$ and so $F.K \geq 0$, which directly contradicts the fact that $F.K = -2$.

This completes the proof of the theorem. □

4.3 Characterization of Geometrically Ruled Surfaces by Extensions of Line Bundles

Geometrically ruled surfaces give us a large family of minimal surfaces (in the case that they are ruled over an irrational curve), and it will turn out that those geometrically ruled over $\mathbb{P}^1$ are almost all minimal (with one exception). In this section, we understand the structure and classification of geometrically ruled surfaces in more depth.

By Corollary 4, any geometrically ruled surface over a curve C can be described as the projectivization of a rank 2 vector bundle over C, defined uniquely up to twisting by a line bundle. Therefore, to classify ruled surfaces, it is important to understand the isomorphism classes of rank 2 vector bundles on a curve. It turns out that over a curve, all rank 2 vector bundles are "built" out of line bundles, which will be made precise in the next lemma.

Lemma 20. Let E be a rank 2 vector bundle on C. We can write E as an extension of M by L, where M and L are both line bundles, meaning there is an exact sequence

$$0 \to L \to E \to M \to 0$$

Proof. First, we identify E with it's sheaf of sections, a locally free $\mathcal{O}_C$ module of rank 2. We take a rational section s of E, locally of the form (f, g) for f, g in the function field of $\mathcal{O}_C$, which may have poles at various points. We can tensor E by a line bundle N, multiplying the transition matrices of E by the $\mathcal{O}_C$ transition function of N to remove these poles. Therefore, $E \otimes N$ has a regular section which we also call s, giving us a map $\varphi : \mathcal{O}_C \to E \otimes N$, sending $a \in \mathcal{O}_C$ to the section $as \in E \otimes N$.

We consider the dual map $\varphi^* : E^* \otimes N^* \to \mathcal{O}_C$. The image is a subsheaf of $\mathcal{O}_C$. Since a subsheaf of a locally free sheaf on a curve C is locally free as well[1], the image of $E^* \otimes N^*$ under φ^* is of the form $\mathcal{O}_C(-D)$. By the same reasoning, the kernel is also a locally free sheaf and must be rank 1, call it M. Then we have an exact sequence of sheaves

$$0 \to M \to E^* \otimes N^* \to \mathcal{O}_C(-D) \to 0,$$

and dualizing gives us an exact sequence

$$0 \to \mathcal{O}_C(D) \to E \otimes N \to M^* \to 0$$

Tensoring by N^* gives us the extension as desired. □

Remark . Note that if E has sections without having to tensor by line bundles N, i.e. $h^0(C, E) \geq 1$, then L can just be taken to be $\mathcal{O}_C(D)$, where D is precisely the divisor of zeros of a section of E. To see this, we just trace through the maps in the above proof. Let (f, g) be the local description of the global section of E, then the map $\mathcal{O}_C \to E$ sends 1 to (f, g). The dual map

[1] This follows form the fact that the image of $E \otimes M^{-1}$ is finitely generated as an $\mathcal{O}_C$ module on trivializations, and that locally freeness can be checked on stalks. The local ring of a curve is a DVR, hence a PID, and the result follows from the PID structure theorem.

$E^* \to \mathcal{O}_C$ sends a local section (a, b) to $af + bg \in \mathcal{O}_C$. The image of the dual map is easily seen to be functions vanish on the common zero set of f and g. In this way, we may take D to be the divisor of zeros of our global section of E.

It turns out that for C not rational, understanding the isomorphism classes of rank 2 vector bundles over C is almost all that is needed to understand isomorphism classes of geometrically ruled surfaces over C. Suppose there is an isomorphism $f : \mathbb{P}_C(E) \cong \mathbb{P}_C(E')$ abstractly as surfaces (not as $\mathbb{P}^1$ bundles). As per the discussion before, we may twist E, E' by line bundles without changing the surfaces $\mathbb{P}_C(E), \mathbb{P}_C(E')$ so that they admit sections. Then we have a morphism α of C given by first applying a section s of $\mathbb{P}_C(E)$, then applying f, then π', so that the following diagram commutes:

$$\begin{tikzcd} \mathbb{P}_C(E) \arrow[r, "f"] \arrow[d, "\pi"] & \mathbb{P}_C(E') \arrow[d, "\pi'"] \\ C \arrow[r, "\alpha"] & C \end{tikzcd}$$

Since C is not rational, by Riemann-Hurwitz f takes fibers of π to fibers of π'. We conclude that α is injective and thus an automorphism of C. Therefore, $\mathbb{P}_C(E)$ and $\mathbb{P}_C(E')$ are isomorphic as surfaces if and only if there is an automorphism α of C such that $\alpha^*(E')$ differs from E by twisting of a line bundle on C.

A natural question to ask is when such an extension of line bundles is trivial, i.e. when can we conclude that E is in fact a direct sum of L and M. We tensor the exact sequence

$$0 \to L \to E \to M \to 0$$

by M^* to get

$$0 \to L \otimes M^* \to E \otimes M^* \xrightarrow{j} \mathcal{O}_C \to 0.$$

This exact sequence splits exactly when we have a map from $\mathcal{O}_C$ module map $i : \mathcal{O}_C \to E \otimes M^*$ such that $j \circ i = \mathrm{id}$. This is exactly asking for the existence of a global section of $E \otimes M^*$ that maps to $1 \in \mathcal{O}_C$. Then taking the long exact sequence, we see that the image of $1 \in H^0(C, \mathcal{O}_C)$ in $H^1(C, L \otimes M^*)$ under the boundary map δ is zero. Since $\delta(1) = 0$ implies the existence of a global section of $E \otimes M^*$ mapping to $1 \in \mathcal{O}_C$, this shows that an extension of line bundles is trivial if and only if this cohomology class is trivial.

This characterization of a trivial extension of line bundles gives us enough information to classify all rank 2 line bundles over $\mathbb{P}^1$, and consequently all geometrically ruled surfaces over $\mathbb{P}^1$

Corollary 5. Every rank 2 vector bundle on $\mathbb{P}^1$ breaks into the direct sum of two line bundles. Therefore, every geometrically ruled surface over $\mathbb{P}^1$ is given by

$$\mathbb{F}_n := \mathbb{P}_{\mathbb{P}^1}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(n))$$

for some n.

Proof. We can first twist E by line bundles to assume that it's degree is 0 or 1. To see this, recall that twisting E by a line bundle doesn't change it's projectivization as discussed in Corollary 4, and that

$$\deg(E \otimes L) := \deg(\det(E \otimes L)) = \deg(L^{\otimes 2} \otimes \det(E)) = 2\deg(L) + \deg(E)$$

For a rank 2 vector bundle E over $\mathbb{P}^1$, writing it as an extension of line bundles L and M gives us that $\chi(E) = \chi(L) + \chi(M)$. Applying Riemann-Roch to L and M gives us that

$$\chi(E) = \deg(L) + \deg(M) - 2g(\mathbb{P}^1) + 2 = \deg(E) + 2$$

This gives us that $h^0(E) \geq 2$, so in particular, E has a global section. By the remark after Lemma 20, for some $k \geq 0$ we can write E as an extension of line bundles

$$0 \to \mathcal{O}_{\mathbb{P}^1}(k) \to E \to \mathcal{O}_{\mathbb{P}^1}(\deg(E) - k) \to 0$$

This extension of $\mathcal{O}_{\mathbb{P}^1}(\deg(E) - k)$ is nontrivial if and only if after tensoring the above exact sequence by $\mathcal{O}_{\mathbb{P}^1}(k - \deg(E))$ and taking long exact sequences, the image of $1 \in H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1})$ is nonzero in $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2k - \deg(E)))$. But the cohomology $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2k - \deg(E)))$ is already trivial, and thus we conclude that every rank 2 vector bundle is of the form $\mathcal{O}_{\mathbb{P}^1}(k) \oplus \mathcal{O}_{\mathbb{P}^1}(\deg(E) - k)$ for some $k \geq 0$. Twisting E by line bundles completes the proof. □

These surfaces $\mathbb{F}_n$ are known as the *Hirzebruch* surfaces. We will explore these surfaces more later in this paper.

In general, we can say more about the kinds of rank 2 vector bundles E we can get from extending one line bundle M by another line bundle L. The extensions E of M by L, i.e. short exact sequences

$$0 \to L \to E \to M \to 0$$

are in fact classified by the image of $1 \in H^0(C, \mathcal{O}_C)$ under $\partial : H^0(C, \mathcal{O}_C) \to H^1(C, L \otimes M^{-1})$ up to scaling by $\mathbb{C}^*$. In this way, it is possible to give a clean description of geometrically ruled surfaces over genus 1 curves, and thus the mimimal models of surfaces birational to $C \times \mathbb{P}^1$ for C an elliptic curve (for more details, see III.15(ii) of [Bea96]). In general, we do not have as explicit of a description of all geometrically ruled surfaces over curves of higher genus.

4.4 The Intersection Theory of Geometrically Ruled Surfaces

We now work to understand the intersection theory on these geometrically ruled surfaces $S = \mathbb{P}_C(E)$ for E a rank 2 vector bundle over C. We have divisors consisting of those pulled back from C, i.e. a finite linear combination of the fibers of the structure map $\pi : \mathbb{P}_C(E) \to C$. We are able to construct one more class of $\mathrm{Pic}(S)$. Consider the pullback π^*E, a rank 2 vector bundle over $\mathbb{P}_C(E)$. By definition, a point $x \in \mathbb{P}_C(E)$ corresponds to a line ℓ_x in $E_{\pi(x)}$, the fiber of E above $\pi(x)$. As the fiber of π^*E at x is naturally identified with the fiber of E at $\pi(x)$, we may consider the sub-line bundle $N \subset \pi^*E$ on $\mathbb{P}_C(E)$ with fiber over x exactly ℓ_x. The line bundle N is the *tautological line bundle* $\mathcal{O}_S(-1)$ on $\mathbb{P}_C(E)$, who's name comes from the fact that if we restrict $\mathcal{O}_S(-1)$ to a fiber $\mathbb{P}^1$ of π, we just get the tautological bundle $\mathcal{O}_{\mathbb{P}^1}(-1)$. See Figure 4.3 for an illustration.

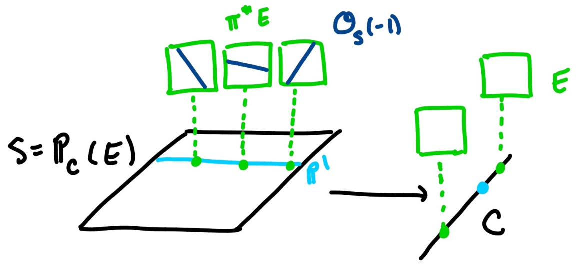

Figure 4.3: An illustration of $\mathcal{O}_S(-1)$

We then define the line bundle $\mathcal{O}_S(1)$ on S by the exact sequence

$$0 \to \mathcal{O}_S(-1) \to \pi^* E \to \mathcal{O}_S(1) \to 0 \tag{4.1}$$

Note that the class of $\mathcal{O}_S(1)$ is linearly independent from the class of the fibers. To see this, we understand $\mathcal{O}_S(1)$ on a fiber: since $\pi^* E$ is trivial over a fiber F of π, by restricting (4.1) to F we get the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^1}(-1) \to \mathbb{P}^1 \times \mathbb{C}^2 \to \mathcal{O}_S(1)|_F \to 0$$

Taking determinants gives us that $\mathcal{O}_{\mathbb{P}^1}(-1) \otimes \mathcal{O}_S(1)|_F$ is the trivial line bundle, so we see that $\mathcal{O}_S(1)|_F$ is just $\mathcal{O}_{\mathbb{P}^1}(1)$! Therefore, since any rational section of $\mathcal{O}_S(1)$ restricts to a rational section of $\mathcal{O}_S(1)|_F$, the intersection number of $c_1(\mathcal{O}_S(1))$ with F is exactly $\deg(\mathcal{O}_S(1)|_F) = 1$. As $F^2 = 0$, we see that $\mathcal{O}_S(1)$ and F are linearly independent in $\mathrm{Pic}(S)$.

As discussed in Lemma 20, we can modify E by twisting by a line bundle so that $h^0(C, E) > 0$, and this twisting doesn't change the surface $\mathbb{P}_C(E)$. Then by the map $H^0(S, \pi^* E) \to H^0(\mathcal{O}_S(1))$ of the long exact sequence of (4.1), we can assume that $\mathcal{O}_S(1)$ has a global section, and we can view $\mathcal{O}_S(1)$ as represented by the divisor h corresponding to the zeros of a generic section. As $h.F = 1$, the projection map $\pi : \mathbb{P}_C(E) \to C$ is a degree 1 map of curves and therefore we see that $h \cong C$.

We know the self-intersection number of a fiber, but what can we say about the self-intersection number of h?

> **Lemma 21.** Let $S = \mathbb{P}_C(E)$ such that E has a global section, and consider $h \in \mathrm{Pic}(S)$ as previously defined. Then we have that
>
> $$h^2 = \deg(E)$$

Note that $\deg(E)$ is *not* invariant under twisting under line bundles, namely that

$$\deg(E \otimes L) = \deg(L^{\otimes 2} \otimes \det(E)) = 2\deg(L) + \deg(E)$$

Therefore, the self-intersection of h is dependent on the choice of presentation E of the surface (which makes sense, as the section h is dependent on E as well).

Proof. We use the interpretation that the self intersection of h is the degree of the normal bundle $\mathcal{O}_S(h)|_h$ of h. To compute this, we consider the exact sequence (4.1) and restrict it to $h \subset S$. Since $\mathcal{O}_S(h) = \mathcal{O}_S(1)$, this gives us the exact sequence

$$0 \to \mathcal{O}_S(-1)|_h \to \pi^*E|_h \to \mathcal{O}_S(h)|_h \to 0$$

Since $\pi : h \to C$ is an isomorphism of curves, we see that the rank 2 bundle $\pi^*E|_h$ over h is isomorphic to rank 2 bundle E over C. If we can show that $\mathcal{O}_S(-1)|_h$ over h is isomorphic to the trivial bundle $\mathcal{O}_C$ over C, then under the isomorphism of h with C our exact sequence would read

$$0 \to \mathcal{O}_C \to E \to \mathcal{O}_S(1)|_h \to 0$$

Therefore, we see that $\det(E) \cong \mathcal{O}_S(h)|_h$ and taking degrees would finish the claim.

To show that $\mathcal{O}_S(-1)|_h$ is trivial over h, we show that there is a nonvanishing section in it. This will crucially use the construction of h, which we got by taking a section s of E over C, pulling it back to a section of π^*E over S, and then pushing it through the map $H^0(S, \pi^*E) \to H^0(\mathcal{O}_S(1))$. We see that s is a nonvanishing section of E: if s vanished on any point of C, then we see that h would contain a whole fiber, contradicting the fact that h intersects every fiber once. Now the subset h of $S = \mathbb{P}_C(E)$ cuts out a sub-line bundle L of E over C by picking a line over each point of C (this uses the fact that h intersects each $\mathbb{P}^1$ fiber once). The fact that the image of s through $H^0(S, \pi^*E) \to H^0(\mathcal{O}_S(1))$ vanishes on h is precisely the statement that s lies inside L (by considering the sequence (4.1) locally as a sequence of vector spaces), so the section s trivializes L over C. Now $\mathcal{O}_S(-1)|_h$ is just the pullback of L under the isomorphism $\pi|_h : h \to C$, completing our claim. □

We claim that the classes of $\mathrm{Pic}(S)$ spanned by the fibers and by h span all of $\mathrm{Pic}(S)$.

Theorem 15. For $S = \mathbb{P}_C(E)$ geometrically ruled over C with structure map π, let h be the class of $\mathcal{O}_S(1)$ in $\mathrm{Pic}(S)$. Then we have that

- $\mathrm{Pic}(S) = \pi^*\mathrm{Pic}(C) \oplus \mathbb{Z}h$
- $H^2(S, \mathbb{Z}) = \mathbb{Z}h \oplus \mathbb{Z}f$, where f is the class of a fiber

Proof. We first show that the map $\pi^*\mathrm{Pic}(C) \oplus \mathbb{Z}h \to \mathrm{Pic}(S)$ sending (π^*P, n) to $\pi^*P + nh$ is an isomorphism. Firstly, it is injective, since if $\pi^*P + nh = 0$, then intersecting with a fiber F (and noting that we can find a linear equivalent divisor to $[p]$ avoiding the support of P) forces $n = 0$, and so $\pi^*P = 0$ as well. To show that this map is surjective, we write any divisor D of S as $D' + nh$ where $D'.F = 0$, and we show that D' is the pullback of a divisor on C (we should hope this since $\pi^*\mathrm{Pic}(S).F = 0$). The strategy will be to show that D' is linearly equivalent to an effective divisor and such an effective divisor must lie in finite collection of fibers of π - while this doesn't work directly, after modifying D' by elements of $\pi^*\mathrm{Pic}(C)$ we are able to carry out our plan.

Consider $D'_m = D' + mF$. Then since $K_S.F = -2$ by the genus formula, we have that $D'_m.K = D.K - 2m$, and we also see that $D'^2_m = D'^2$. Finally, we also have that $h^0(K - D'_m) = 0$ for large enough m. To get this, note that since S is projective, we can consider a hyperplane section H. Now $H.F > 0$ since we may pick a hyperplane not containing F. If $h^0(K-D'_m) > 0$ for all m, then $H.(K-D'_m) = H.(K-D')-m(H.F)$ must be positive for all m, which cannot be true by our previous observation. Now applying Riemann-Roch for surfaces gives us that

$$h^0(D'_n) \geq \chi(O_S) + \frac{1}{2}(D'^2 - D.K + 2n)$$

Since the right side increases in n, we may pick n large enough so that there exists an effective divisor E linearly equivalent to D'_n. But $\pi(E)$ is a closed set of C. If $\pi(E)$ isn't a finite set of points, $\pi(E)$ must be all of C, and thus $E.F > 0$, a contradiction to the fact that $D'_m.F = 0$. Therefore E must lie in the support of a finite set of fibers, thus must be a linear combination of fibers, so is an element of $\pi^*\mathrm{Pic}(C)$. Since mF is clearly in $\pi^*\mathrm{Pic}(C)$ for all m, we conclude that $D' \in \mathrm{Pic}(C)$ as well. This concludes the proof of the first bullet point.

Now we prove the second bullet point. In Lemma 18, we showed for geometrically ruled surfaces that the Néron-Severi group is in fact equal to $H^2(S, \mathbb{Z})$. As F and $c_1(O_S(1))$ are linearly independent as homology elements (for the same reason they are linearly independent as classes in Pic: $[F]^2 = 0$ yet $Q_S([F], c_1(O_S(1))) = 1$), and since all fibers of π are homologous, we conclude that $H^2(S, \mathbb{Z}) = \mathbb{Z}h \oplus \mathbb{Z}f$. □

From this intersection theory, we can show that the Hirzebruch surfaces are all distinct from each other, and characterize those that are minimal:

Corollary 6. All of the $\mathbb{F}_n$ are distinct, and all are minimal except for $n = 1$, which is $\mathbb{P}^2$ blown up at a point.

Proof. We will show that for $n, m \geq 1$, $\mathbb{F}_n$ and $\mathbb{F}_m$ are distinct for $n \neq m$ by showing that $\mathbb{F}_n$ has a unique curve of negative self-intersection and its self-intersection number is $-n$. We construct this curve, which we call N, as follows: we give a presentation of $\mathbb{F}_n$ by the rank 2 line bundle $E = O_{\mathbb{P}^1} \oplus O_{\mathbb{P}^1}(n)$, so that with h in $\mathrm{Pic}(\mathbb{F}_n)$ as defined before, $h^2 = n$. We consider $L = \{0\} \oplus O_{\mathbb{P}^1}(n)$ as a sub-line bundle of E, which cuts out curve N of $\mathbb{F}_n = \mathbb{P}_{\mathbb{P}^1}(E)$. Clearly $N.F = 1$ since N is a section of the structure map $\pi : \mathbb{P}_{\mathbb{P}^1}(E) \to \mathbb{P}^1$. Since the pullback of L under $\pi|_N : N \to C$ is exactly $O_{\mathbb{F}_n}(-1)|_N$, we see that $\deg(O_{\mathbb{F}_n}(-1)|_N) = n$. Since π^*E restricted to N has degree n as well, we see (by taking determinants of (4.1) after restricting to N) that $O_S(1)|_N$ must have degree 0. We conclude that $h.N = \deg(O_S(h)|_N) = 0$. Writing N as $aF + bh$, dotting with F gives $b = 1$, and dotting with h gives $a = -n$. Thus $N = h - nF$, and $N^2 = -n$.

Now we show no other curve in $\mathbb{F}_n$ has negative self-intersection. Let C be any curve in $\mathbb{F}_n$ not equal to N. It's class in $\mathrm{Pic}(\mathbb{F}_n)$ is of the form $aF + bh$. Since C is effective (and $F^2 = 0$), $C.F \geq 0$ and b is nonnegative. Since $C.N \geq 0$ (since $C \neq N$ and both are effective), a is nonnegative as well, so $C^2 \geq 0$. Therefore, we've shown that $\mathbb{F}_n$ and $\mathbb{F}_m$ are distinct for $n \neq m$, $n, m \geq 1$. $\mathbb{F}_0$ is just $\mathbb{P}^1 \times \mathbb{P}^1$ and has no curves of negative self-intersection, and thus is distinct from the others.

From Castelnuovo's Contractability Criterion it follows immediately that $\mathbb{F}_n$ is minimal for $n \neq 1$. To show that $\mathbb{F}_1$ is $\mathbb{P}^2$ blown up at a point, note that viewing the blowup of $\mathbb{P}^2$ as a subset of $\mathbb{P}^2 \times \mathbb{P}^1$, the blowup map ε is the projection onto the $\mathbb{P}^2$ coordinate. Then the projection π onto the $\mathbb{P}^1$ coordinate gives $\mathrm{Bl}_{[0:0:1]}(\mathbb{P}^2)$ the structure of a geometrically ruled surface containing a curve of self intersection -1, thus it must be $\mathbb{F}_1$. $\square$

4.5 Minimal Models of Rational Surfaces

Our argument in Theorem 14 to show that all minimal models of irrational ruled surfaces (those birational to $C \times \mathbb{P}^1$ for C irrational) are geometrically ruled does not work for C rational, as we frequently make use of the fact that the image of a rational curve (the exceptional divisor of a blowup in our case) into C must be a point. To this end, we try listing all of the minimal rational surfaces we know so far:

- The Hirzebruch surfaces $\mathbb{F}_n$ for $n \neq 1$ (since $\mathbb{P}^1 \times \mathbb{P}^1$ is birational to $\mathbb{P}^2$)
- $\mathbb{P}^2$

It turns out that there are all of the rational minimal surfaces that we can have!

How do we plan to show such a result? Consider a minimal rational S that is a geometrically ruled surface, i.e. a $\mathbb{P}^1$ fibration over $\mathbb{P}^1$. The data of this fibration is exactly captured by the existence of a particular linear pencil on S: S being geometrically ruled over $\mathbb{P}^1$ is equivalent to showing the existence of a linear system $|C|$ of dimension 2 without basepoints consisting solely of rational curves. If we can find such a linear pencil on our surface S, we would get a map to $\mathbb{P}^1$ with rational fibers, which by Noether-Enriques and Corollary 4 would allow us to conclude that S is geometrically ruled over $\mathbb{P}^1$ and isomorphic to one of the $\mathbb{F}_n$ for $n \neq 1$.

Now suppose that our minimal rational surface S was just $\mathbb{P}^2$. Then an isomorphism $S \to \mathbb{P}^2$ is the same as the data of a linear system $|C|$ on S with $h^0(S, \mathcal{O}_S(C)) = 3$ where every pair of divisors in $|C|$ intersect exactly once. To see this, let a basis of $h^0(S, \mathcal{O}_S(C))$ be given by the sections s_0, s_1, s_2, so the map $\varphi : S \to \mathbb{P}^2$ induced by $|C|$ is defined by sending a $p \in \mathbb{P}^2$ to

$$p \mapsto [s_0(p) : s_1(p) : s_2(p)]$$

Now the fiber of $[a : b : c] \in \mathbb{P}^2$ under φ are exactly those p such that

$$bs_0(p) - as_1(p) = 0 \text{ and } cs_1(p) - bs_2(p) = 0$$

Therefore, the fiber of a point in $\mathbb{P}^2$ is the intersection of the zero set of the distinct divisors defined by the sections $bs_0 - as_1$ and $cs_1 - bs_2$. Since we want the fibers of φ to have a single element with multiplicty 1 and are assuming S is $\mathbb{P}^2$, we need by Bezout's theorem that all divisors obtained by the zero sets of $bs_0 - as_1$ and $cs_1 - bs_2$ must be degree 1 curves in $\mathbb{P}^2$, i.e. rational.

In both cases, it is crucial that for a minimal rational surface S we show the existence of a rational curve C inside of it with self-intersection 0 or 1. The crux of this claim will rely on the following techincal lemma:

> **Lemma 22.** Let S be a minimal surface with $h^1(S, \mathcal{O}_S) = h^0(S, \mathcal{O}_S(2K)) = 0$. Then there is an effective divisor D on S with both $K.D < 0$ and $|K + D| = \emptyset$.

We will not present the proof in this exposition, instead referring to the reader to Chapter V.9 of [Bea96] for a proof of this lemma. We will show however how this lemma implies the existence of a curve with positive self-intersection:

Corollary 7. Under the assumptions of Lemma 22, there is a rational curve C with $C^2 \geq 0$.

Proof. We let D be as in the result of Lemma 22. Writing $D = \sum n_i C_i$, there is some C_i such that $K.C < 0$. We still have that $|K+C| = \emptyset$ (if it wasn't, $|K+D|$ wouldn't be empty either), so Riemann-Roch to $K + C$ gives us that

$$\chi(K + C) = \chi(O_S) + \frac{1}{2}(K^2 + C.K)$$

But using the fact that $h^0(S, O_S(K + C)) = 0$ and $h^2(S, O_S(K + C)) = h^0(S, O_S(-C)) = 0$ by Serre duality, as well as that $h^0(S, O_S) = 1$, $h^1(S, O_S) = 0$, we conclude that

$$0 = h^0(S, O_S(K + C)) \geq 1 + \frac{1}{2}(K^2 + C.K) = g(C)$$

where the last equality is the adjunction formula. But $h^0(C)$ is 1 as well, so C is a rational curve, thus smooth. Since we assumed that $K.C < 0$, adjunction again gives us that $C^2 \geq -1$, but since S is minimal we conclude C has nonnegative self-intersection. □

Note that rational surfaces satisfy $h^1(S, O_S) = 0$ by Lemma 17, since $\mathbb{P}^2$ is birational to $\mathbb{P}^1 \times \mathbb{P}^1$. Rational surfaces also satisfy $h^0(S, O_S(2K)) = 0$. This is because $h^0(S, O_S(2K))$ is a birational invariant by the same argument as Lemma 17, and we can compute this for $\mathbb{P}^2$: adjunction gives that for a line H, $K.H = -3$, so $K = -3H$, $O_{\mathbb{P}^2}(2K) = O_{\mathbb{P}^2}(-6)$, and so $h^0(\mathbb{P}^2, O_{\mathbb{P}^2}(2K)) = 0$.

With this, we can attack what we set out to prove.

Theorem 16. Every minimal rational surface S is isomorphic to $\mathbb{P}^2$ or one of the Hirzebruch surfaces $\mathbb{F}_n$ for $n \neq 1$.

Proof. Fix a hyperplane section H of S. By our previous Lemma 22, we may find a curve of nonnegative self intersection on S. Choose a C that minimizes it's self intersection over all curves in S, and from those of minimal self-intersection choose one that minimizes it's intersection with H.

We first need to show that any divisor D in $|C|$ is actually a smooth rational curve, as we discussed was necessary in the beginning of this section. To see this, write D as $\sum_{i=1}^{k} n_i C_i$, and we will show that D is actually one of these curves. (We should hope, from our discussion in the beginning of this subsection, that these have nonnegative self-intersection). First of all, note that all of these C_i are in fact smooth rational curves. To see this, we use the genus formula: we know that $(K + C).C = -2$, and therefore $h^0(S, O_S(K + C)) = 0$[2]. This immediately tells us that $|K + C_i| = \emptyset$ for all i, which by the methods of Corollary 7 allow us to conclude the rationality of the C_is.

[2] Otherwise, there would be an effective divisor D such that $D.C = -2$. Writing $D = nC + D'$ for D' effective and disjoint from S gives that $D.C > 0$ (using crucially that $C^2 \geq 0$), a contradiction.

Now since $K.C < 0$, there must be some C_i where $K.C_i < 0$ as well. Since $(K + C).C = -2$, we must have $C_i^2 \geq -1$, but by the minimality of S we have the even stronger $C_i \geq 0$. We now claim that $D = C_i$: writing $D' = D - n_i C_i$, we have that

$$C^2 = n_i^2 C_i^2 + n_i C_i.D' + D.D'$$

Clearly $D.D' \geq 0$ since $C.D' \geq 0$, and the same is true of $C_i.D'$ as C_i, C_j are disjoint for all $i \neq j$. Therefore, $C^2 \geq n_i^2 C_i^2 \geq 0$, and by the minimality of C^2 we get that $C_i^2 = C^2$ and $n_i = 1$. Now intersecting C with a hyperplane section H, we get that $H.D' = 0$ by the minimality of $C.H$ over all curves with self intersection equal to that of C. But clearly $H.D' = 0$ implies that $D' = 0$, proving that every divisor D in $|C|$ is in fact smooth rational.

We next show that $h^0(|C|) \leq 3$, which would force the maps given by these linear systems to be into $\mathbb{P}^1$ or $\mathbb{P}^2$, the only two possibilities we would hope to see. This follows easily: pick a point p not on C. The vector space of functions in a trivialization of $\mathcal{O}_S(C)$ that vanish to at most order 1 at a fixed point p is dimension 3 over $\mathbb{C}$ (since the maximal ideal of the local ring at p is generated by two elements). If $h^0(|C|) \geq 4$, then a divisor in $|C|$ would vanish to order 2 at p, and thus wouldn't be smooth, a contradiction to the fact that all divisors in $|C|$ are smooth rational curves.

Finally, we construct our linear system from the residue exact sequence

$$0 \to \mathcal{O}_S \to \mathcal{O}_S(C) \to \mathcal{O}_C(C) \to 0 \tag{4.2}$$

We note that $\mathcal{O}_C(C)$ is just $\mathcal{O}_{\mathbb{P}^1}(C^2)$. Since the line bundle $\mathcal{O}_C(C) = \mathcal{O}_{\mathbb{P}^1}(C^2)$ is basepoint-free (since $C^2 \geq 0$) and $h^0(C, \mathcal{O}_C)$ is just the restriction of the linear system $|C|$ to C, we see that $|C|$ is basepoint-free on C and is thus basepoint-free on all of S! Furthermore, since $h^0(S, \mathcal{O}_S) = 0$ by the fact that S is rational, (4.2) is exact after taking global sections as well, forcing $h^0(S, \mathcal{O}_S(C)) = C^2 + 2$. Since we just showed $h^0(C, \mathcal{O}_S(C)) \leq 3$, we must have that either:

- $C^2 = 0$, in which case $|C|$ gives a morphism from S to $\mathbb{P}^1$ with smooth rational fibers, and Noether-Enriques lets us conclude that S is geometrically ruled over $\mathbb{P}^1$, or
- $C^2 = 1$, in which case $|C|$ gives us a morphism φ from S to $\mathbb{P}^2$ with fibers the intersection of two distinct divisors in $|C|$. Since $C^2 = 1$, we conclude that every fiber of φ is a single point of multiplicity 1, thus φ is a finite map of degree 1. But a finite degree 1 map of projective varieties is necessarily an isomorphism, and S must be $\mathbb{P}^2$!

$\square$

Chapter 5

All the Other Surfaces

Now that we have understood the minimal models of ruled surfaces and rational surfaces, we work to conclude the classification of minimal models of smooth surfaces by showing the following result:

> **Theorem 17.** Let S be a smooth complex projective surface that is not ruled or rational. Then S has a unique minimal model.

To motivate the proof of this theorem, we explore what possible obstructions we could have to a unique minimal model. Suppose that we have a birational map φ of surfaces $S' \dashrightarrow S$ that is not an isomorphism, which by our understanding of birational geometry in Theorem 10 factors as

$$\begin{tikzcd} & \hat{S} \arrow[dl, "f"'] \arrow[dr, "g"] & \\ S' \arrow[rr, dashed, "\varphi"] & & S \end{tikzcd}$$

where both f and g are compositions of blowups. We pick $f = \pi_n \circ \ldots \circ \pi_1$ a sequence of blowups such that n is minimal, and we may assume that $n \geq 1$ (otherwise, flip the direction of the map φ, since f and g cannot both be isomorphisms). By Lemma 16, g sends the exceptional divisor E_n of the last blowup π_n to a curve C in S'. Since E_n is rational and g is nonconstant as a map $E_n \to C$, C must be rational as well by Riemann-Hurwitz. Since we understand the genus of C, we hope to get a handle on $K_S.C$ to extract information of the self-intersection of C via adjunction.

We write $g : \widehat{S} \to S$ as the composition of blowups ε_i:

$$\hat{S} \xrightarrow[\varepsilon_n]{} S_{n-1} \xrightarrow[\varepsilon_{n-1}]{} S_{n-2} \xrightarrow[\varepsilon_{n-2}]{} \cdots \xrightarrow[\varepsilon_2]{} S_1 \xrightarrow[\varepsilon_1]{} S$$

We observe that the strict transform of C under g (in the sense that we take the strict transform of C under the map $S_1 \to S$, then the strict transform of this under the map $S_2 \to S_1$, and so on) is just E_n. To see this, note that $g^{-1}(C)$ as a divisor must be supported on E_n and any of the exceptional divisors of the blowups in g. As C is irreducible (since E_n is), the strict transform of C under g must be irreducible as well, and therefore we see it must be E_n. We know that $K_{\widehat{S}}.E_n = -1$, and to relate $K_S.C$ to $K_{\widehat{S}}.E_n$ it is enough to understand how $K_S.C$ changes blowup by blowup.

Consider the first blowup $\varepsilon_1 : S_1 \longrightarrow S$, and E the exceptional divisor of ε_1. We have that

$$K_S.C = \varepsilon_1^*(K_S).\varepsilon_1^*(C) = (K_{S_1} - E)(\widehat{C} + mE)$$

where $\widehat{C}$ is the strict transform of C in S_1, and m is the multiplicity that C intersects the blown up point. Then we calculate

$$(K_{S_1} - E)(\widehat{C} + mE) = K_{S_1}.\widehat{C} - m - m + m = K_{S_1}.\widehat{C} - m,$$

and we conclude that $K_S.C \leq K_{S_1}.\widehat{C}$, with equality happening when the point we blow up at is not on C. By induction, we conclude that

$$K_S.C \leq K_{\widehat{S}}.E_n = -1$$

This inequality is an equality precisely when the function g consists of blowups at points disjoint from C. However, if $K_S.C = -1$, then since C is rational we would get that $C^2 = -1$ by adjunction, a contradiction to the minimality of S. Thus we must have that $K_S.C \leq -2$. By adjunction again on C, we conclude the following result:

> **Lemma 23.** If a minimal surface S is not unique in it's birational equivalence class, then it contains a rational curve C such that $C^2 \geq 0$.

The existence of such a rational curve with nonnegative self-intersection puts very strong constraints on the geometry of our surface, and will be enough to conclude that S is ruled or rational as we will see in the following arguments. If a surface happens to be ruled, i.e. birational to $C \times \mathbb{P}^1$, we know from Lemma 17 that it's first cohomology with respect to the structure sheaf is just the genus of C. Therefore, if $h^1(S, \mathcal{O}_S) = 0$, we hope to build a map from S to $\mathbb{P}^1$ via a linear system, and conclude by Noether-Enriques that S is birational to a geometrically ruled surface over $\mathbb{P}^1$. In the case of $h^1(S, \mathcal{O}_S) > 0$, we instead need to construct a map from S to a curve of positive genus, which the machinery of linear systems does not help us with. To achieve this task, we will introduce the machinery of the Albanese torus.

5.1 The Case Where $h^1(S, \mathcal{O}_S) = 0$

We first prove Theorem 17 in the case where $h^1(S, \mathcal{O}_S) = 0$. Suppose that S does not have a unique minimal model, then by Lemma 23 of the previous section there is a rational curve C in S such that $C^2 \geq 0$. What the condition $h^1(S, \mathcal{O}_S) = 0$ will allow us to do is guarantee that C has space to move and allow us to construct a pencil of curves. Explicitly, we again consider the residue exact sequence

$$0 \to \mathcal{O}_S \to \mathcal{O}_S(C) \to \mathcal{O}_C(C) \to 0$$

Since $H^1(S, \mathcal{O}_S) = 0$, taking long exact sequences gives us that

$$h^0(S, \mathcal{O}_S(C)) = h^0(S, \mathcal{O}_S) + h^0(C, \mathcal{O}_C(C))$$

The fact that C is rational and $\deg(\mathcal{O}_C(C)) = C^2 \geq 0$ lets us conclude that $h^0(C, \mathcal{O}_C(C)) \geq 1$ and $h^0(S, \mathcal{O}_S(C)) \geq 2$. Therefore, C is linearly equivalent to an effective divisor D not sharing any common components with C, and together they define a rational map $\varphi : S \dashrightarrow \mathbb{P}^1$ with basepoints at the points where C and D intersect. After blowing up S at these points, we get a well-defined morphism $\widetilde{\varphi} : \widehat{S} \to \mathbb{P}^1$, with the fiber over $[1:0]$ equal to the strict transform of C, which is still a smooth rational curve. Since $\widetilde{\varphi}$ is a surjective map from a smooth surface to a smooth curve, we can apply Noether-Enriques to conclude that S is a ruled surface.

5.2 The Case Where $h^1(S, \mathcal{O}_S) > 0$, and the Albanese Torus

To prove Theorem 17 in the case of $h^1(S, \mathcal{O}_S) > 0$, we need a way to get a map not to $\mathbb{P}^1$ but to a curve of positive genus. To do this, we need to introduce the Albanese torus, the analog of the Jacobian of a curve for higher-dimensional varieties.

Let X be a compact complex projective manifold of any dimension. For any holomorphic 1-form $\omega \in \Omega^1_X$ and a homology class $\alpha \in H^1(X)$, we have a *period map* $\varphi_\alpha \in H^0(X, \Omega^1_X)^*$ defined by

$$\omega \mapsto \int_\alpha \omega \tag{5.1}$$

The space of period maps, denoted by Λ, form a lattice inside $H^0(X, \Omega^1_X)^*$. To see this, first note that by the Dolbeault theorem, $H^0(X, \Omega^1_X)^* = H^{1,0}(X)^*$. Consider the map $H_1(X, \mathbb{R}) \to H^0(X, \Omega^1_X)^*$ given by evaluation (just (5.1) but α is a real homology class). If the image of a homology class $[\alpha]$ is zero, under the isomorphism $H^{1,0}(X) \cong \overline{H^{0,1}}(X)$ we see that α evaluates to zero on every complex 1-form and therefore $[\alpha] = 0$ by Poincaré duality. Since $H_1(X, \mathbb{R}) \to H^0(X, \Omega^1_S)^*$ is a map of $\mathbb{R}$-vector spaces of the same dimension (both $b_1(X)$), we conclude the map $H_1(X, \mathbb{R}) \to H^0(X, \Omega^1_S)^*$ is an isomorphism, and clearly $H_1(X, \mathbb{Z})$ forms a lattice in $H_1(X, \mathbb{R})$.

Definition 10. The *Albanese torus* $\mathrm{Alb}(X)$ of a compact complex projective manifold is defined to be the complex torus

$$\mathrm{Alb}(X) = H^0(X, \Omega^1_X)^* / H_1(X, \mathbb{Z})$$

The Albanese torus has dimension $h^0(X, \Omega^1_X)$.

For smooth curves, it is clear that the Albanese torus is just the Jacobian.

For any variety X, we have the *Abel-Jacobi map* $\alpha : X \to \mathrm{Alb}(X)$ with basepoint $p \in X$ defined by

$$\alpha(x) = \left(\omega \mapsto \int_{\gamma_{p,x}} \omega\right)$$

where $\gamma_{p,x}$ is a path from p to x. Changing the path $\gamma_{p,x}$ changes the image of α by a period, and thus α is well defined. Changing the basepoint of the map induces a translation on $\mathrm{Alb}(X)$. We now show this map is holomorphic. If $\omega_1, \ldots, \omega_n$ is a basis of $H^0(X, \Omega_X^1)$, then asking for the map $\alpha : X \to \mathrm{Alb}(X)$ to be holomorphic is asking for the map $\mathbb{C}^{\dim(X)} \to \mathbb{C}^{h^0(X,\Omega_X^1)}$ given by

$$x \mapsto \left(\int_{\gamma_{p,x}} \omega_i \right)_{1 \le i \le n}$$

to be holomorphic in x. Since changing p only changes the above map by a constant and being holomorphic is a local statement, we may take x to be in a disk centered at p where ω_i is locally described by a holomorphic function. From here, the holomorphicity of α follows from the fundamental theorem of calculus.

The Albanese torus $\mathrm{Alb}(X)$ is the "universal" complex torus corresponding to X in the following sense:

Theorem 18. (Universal Property of the Albanese torus) For any complex torus T and a map $f : X \to T$, there is a unique map f' such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\alpha"] \arrow[dr, "f"'] & \mathrm{Alb}(X) \arrow[d, "\exists! f'"] \\ & T \end{tikzcd}$$

We refer the reader to V.13 of [Bea96] for a proof.

Remark . We will use the fact that the Albanese torus is projective. This can be proven using the Kodaira embedding theorem, though we will not present it here.

Immediately from the universal property, we conclude that the Albanese torus is functorial: for f a map $X \to Y$, the universal property on the map $X \to Y \to \mathrm{Alb}(Y)$ factors through $\mathrm{Alb}(X)$ and so the following diagram commutes.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\alpha_X"'] & Y \arrow[d, "\alpha_Y"] \\ \mathrm{Alb}(X) \arrow[r, "F"'] & \mathrm{Alb}(Y) \end{tikzcd}$$

We now show that in the setting of $h^1(S, \mathcal{O}_S) \ge 1$ that we are interested in, the Albanese torus provides us a way to construct maps from a surface to a smooth curve of positive genus.

Lemma 24. Let S be a surface with $h^1(S, \mathcal{O}_S) \ge 1$. Then the image of $\alpha : S \to \mathrm{Alb}(S)$ is a smooth curve of genus $h^1(S, \mathcal{O}_S)$.

Proof. Since this map is of projective varieties we know the image of α is either a point, curve, or a surface. We see immediately that α can't be a point - the Albanese torus of X has dimension at least 1, and if $\alpha(X)$ were a point, $\alpha(S)$ would satisfy the universal property of $\mathrm{Alb}(S)$ as well. Now suppose that $\alpha(S)$ is a surface. Then we may find an open set U of $\alpha(S)$ where α is smooth, and as the fibers are 0-dimensional over U, $\alpha^{-1}(U)$ is a covering space over U in the analytic topology. Pick local coordinates $u_1, \ldots, u_n$ of $\mathrm{Alb}(X)$ at a point $p \in U$ where $\alpha(S)$ is defined by $u_3 = u_4 = \ldots = u_n = 0$. Then $du_1 \wedge du_2$ is a nonzero 2-form on $\alpha(S)$. But since the charts of complex tori only differ by translation, $du_1 \wedge du_2$

defines a global 2-form on $\mathrm{Alb}(S)$. Pulling back this 2-form gives a 2-form on S that is nonvanishing at any point in $\alpha^{-1}(p)$ by construction, a contradiction. Therefore, $\alpha(S)$ is a curve C.

We now show that C is smooth. For the sake of contradiction, suppose it is not. Then we may consider it's normalization $\widetilde{C}$, which for curves is smooth. By the universal property of normality, since S is smooth the Abel-Jacobi map factors through $\widetilde{C}$. From both functoriality and the universal property of the Albanese torus, we have the commuting diagram

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "\alpha_S"'] & \widetilde{C} \arrow[d, "\alpha_{\widetilde{C}}"'] \arrow[dr, "g"] & \\ \mathrm{Alb}(S) \arrow[r, "F"'] & \mathrm{Jac}(\widetilde{C}) \arrow[r, "G"'] & \mathrm{Alb}(S) \end{tikzcd}$$

Since the map f is surjective, so is the map F[1]. Since $G \circ F$ must be the identity map by the universal property of Albanese tori, we conclude that G and F are isomorphisms of $\mathrm{Alb}(S)$ and $\mathrm{Jac}(\widetilde{C})$ that are inverse to each other. Since the map from a smooth curve to it's Jacobian is an embedding, $G \circ \alpha_{\widetilde{C}}$ is an embedding, and thus g is as well, so we conclude that $C = \widetilde{C}$. Since $\mathrm{Jac}(\widetilde{C}) \cong \mathrm{Alb}(S)$, we see that $\widetilde{C}$ has genus $h^0(X, \Omega_X^1)$ and so C does as well. □

We are now able ready to conclude the proof of Theorem 17 in the case that $h^1(S, \mathcal{O}_S) > 0$. By the previous lemma, the image of $\alpha : S \to \mathrm{Alb}(S)$ is a smooth curve B of positive genus.

We first show that the fibers of α are connected using Stein Factorization (Theorem 4): we factor α as $S \to B' \to B$. We can take the normalization B'' of B', and by the universal property of normalization $S \to B'$ factors through B''. Now $S \to B''$ has connected fibers since $S \to B'$ does, and B'' is smooth since it's the normalization of a curve. Thus, without loss of generality we may assume that B' is smooth. Now we have by functoriality the commuting diagram

$$\begin{tikzcd} B' \arrow[r] \arrow[d, "\alpha_{C'}"'] & B \arrow[d, "\alpha_C"'] \\ \mathrm{Alb}(B') \arrow[r] & \mathrm{Alb}(B) \end{tikzcd}$$

and the map $\mathrm{Alb}(B') \to \mathrm{Alb}(B)$ is an isomorphism by the same argument showing the maps F and G are inverses in Lemma 24. Since both $\alpha_{B'}$ and α_B are embeddings, so is the map $B' \to B$. Therefore $B \cong B'$ and the fibers of α are connected.

Recall that by Lemma 23, if S does not have a unique minimal model then there is a rational curve C with nonnegative self-intersection in S. Since B has genus greater than 1, C maps to a point under α, and C must lie in a fiber F of α. If F is of the form $\sum_{i=1}^k n_i C_i$ for $k > 1$, then $C_i^2 < 0$ for all i (by the same computation at the end of Theorem 14), which can't happen since $C^2 \geq 0$. Thus $F = rC$ for some r.

With this we show that a generic fiber of α is a $\mathbb{P}^1$. Consider a point p in B such that α is smooth, so that the fiber F' of p is a smooth irreducible curve (since the fibers of α are connected). By the genus

[1]To see this, note that the abelian subvariety of $\mathrm{Alb}(S)$ generated by $\alpha_S(S)$ must be all of $\mathrm{Alb}(S)$ (or else $\alpha_S(S)$ satisfies the universal property of the Albanese variety as well), and the same is true for $\widetilde{C}$. But since f is surjective, the image of F is precisely the subvariety of $\mathrm{Jac}(\widetilde{C})$ generated by the image of $\alpha_{\widetilde{C}}$.

formula, $2g(F') - 2 = F'^2 + F'.K_S$. But F' and F are numerically equivalent and $F.K_S = -2r$ by the genus formula on C, so we conclude that

$$-2r = 2g(F') - 2$$

This forces $r = 1$ and $g(F') = 0$, so that the map $\alpha : S \to B$ has fiber over p a smooth $\mathbb{P}^1$. We conclude by Noether-Enriques that S is ruled.

Chapter 6

The Minimal Model Program

Over the course of the past several chapters, we showed that outside of ruled and rational surfaces, every surface has a unique minimal model. For the ruled and rational surfaces, the minimal surfaces that arise are exactly $\mathbb{P}^2$ and projectivized rank 2 vector bundle on a smooth curve.

However, our analysis for both of these cases felt very disjointed. In the case of ruled and rational surfaces, the fact that (outside of $\mathbb{P}^2$) our minimal models were fibrations over a curve came from a strong understanding of the geometry these particular surfaces. In the case of non-ruled surfaces, the fact that our minimal models were unique came from a study of rational curves of nonnegative self-intersection on S. In this last section, we we will reframe these arguments in the lens of the more general framework of the *Minimal Model Program*, a direction of research that studies the birational geometry of algebraic varieties by finding "nice" representatives of each birational equivalence class. Through this process, we will see that the structural results we obtained for ruled and non-ruled surfaces come from a unifying geometric principal. We will also discuss the extent to which the Minimal Model Program for surfaces extends to higher dimensional complex projective varieties. In the case of smooth surfaces, we were able to find a smooth "minimal" representative in every birational equivalence class, i.e. a smooth surface that is not the blowup of another smooth surface. In higher dimensions, the program becomes much more complicated, and we must change our definition of "minimal."

This section attempts connect our study of surfaces to the program and give a glimpse of the vast landscape of tools and approaches to the birational classification problem in higher dimensions. For substantially more detailed expositions of the Minimal Model Program, see for instance the survey articles [Kol87], [Cad+05], and [Mor24].

6.1 An Overview of the Program

What is a good definition of a "minimal model" of a birational equivalence class in higher dimensions? For a surface, we were able to show that for every birational morphism $S \to S'$ of smooth surfaces, S is isomorphic to S' blown up some number of times. Thus, we were able to define a minimal surface S as one such that every birational morphism from it is an isomorphism, or equivalently, that S is not the blowup of any other surface.

In arbitrary dimensions, we can try doing the same thing. While we won't define it rigorously here, we can perform blowups of a general closed subvariety by replacing the closed subvariety with the "space of normal directions to the closed subvariety". See for example Chapter 2.5 of [Huy05] for a description in the complex analytic setting. Analogous to the resolution of indeterminacy for surfaces, we have the following theorem of Hironaka. This is a much, much harder theorem than in the surface case.

Theorem 19. (Elimination of Indeterminacy, characteristic zero) Let X be a smooth projective variety over $\mathbb{C}$ (more generally any characteristic zero algebraically closed field), Z a projective variety, and $\varphi : X \dashrightarrow Z$ a rational map. Then there exists a sequence of blowups $\varepsilon_i : X_i \longrightarrow X_{i-1}$ along closed smooth subvarieties of X_{i-1} such that $f = \varphi \circ \varepsilon_1 \circ \ldots \circ \varepsilon_n$, i.e. the following diagram commutes:

$$\begin{tikzcd}
X \arrow[drrrr, dashed, "\varphi"'] & X_1 \arrow[l, "\varepsilon_1"'] & \cdots \arrow[l, "\varepsilon_2"'] & & X_n \arrow[ll, "\varepsilon_n"'] \arrow[d, "f"] \\
& & & & Z
\end{tikzcd}$$

So we could try to define a minimal smooth n-dimensional projective variety as one that is not a blowup of any other smooth n dimensional projective variety along a smooth subvariety. While this is a perfectly valid definition, historically this turned out to be a very difficult definition to work with and very little progress was made in this direction. In the early 1980s, work of Reid and Mori highlighted a key shift in the "correct" definition of minimality.

Firstly, we call a divisor $D \in \mathrm{Pic}(X)$ *numerically effective*, or *nef* for short, if for every curve C, $(D.C) \geq 0$. We now give a new definition of minimality:

Definition 11. A smooth projective variety X is *minimal* if it's canonical bundle K_X is numerically effective.

Here is some motivation for this definition: Our current procedure for constructing minimal surfaces from a smooth surface S is to contract rational curves of self-intersection -1 (or -1 curves for short) iteratively, which must terminate since every time we do so the rank of $NS(S)$ drops by one. Each of these -1 curves C has intersection with the canonical class $K_S.C = -1$, and so form an obstruction to the canonical class being numerically effective. Our new definition of minimal captures all obstructions to K_S being nef, not just the -1 curves considered before.

In fact, for non ruled and rational surfaces, this definition of minimality agrees with our previous definition for surface, as we will see in a moment. Therefore, it should be the case that for non ruled and rational surfaces S, there is a unique representative of the birational equivalence class of S with K_S nef. This follows immediately from the following Lemma, which is just the argument from the beginning of Chapter 5.

Lemma 25. Suppose that S is a minimal and K_S is numerically effective. Then every birational map $S' \dashrightarrow S$ is in fact a morphism.

Proof. We have the diagram

$$\begin{tikzcd} & \hat{S} \arrow[dl, "\varepsilon"'] \arrow[dr, "\pi"] & \\ S' \arrow[rr, dashed, "\varphi"'] & & S \end{tikzcd}$$

Where ε and π are a sequence of blowups and isomorphisms. Let E be the exceptional divisor corresponding to last blowup ε_n of $\varepsilon = \varepsilon_1 \circ \cdots \circ \varepsilon_n$, and we will show that E must be contracted to a point, thus by applying Lemma 16 inductively, φ must be a morphism.

Suppose for contradiction that $\pi(E)$ is a curve in S, then by checking blowup by blowup as we did in the beginning of Section 5 we have that

$$K_S.C \leq K_{\widehat{S}}.E$$

But $K_{\widehat{S}}.E = -1$ by adjunction, so $K_S.C \leq -1$. This contradicts the fact that K_S is numerically effective. Therefore, φ is a morphism. □

By replacing our original definition of minimality for surfaces with that of Definition 11, the goal shifts from contracting -1 curves to contracting *all* curves that intersect K_S negatively. This leaves several questions to be explored, but two that stand out are

1. What other curves are there that form obstructions to K_S being nef?

2. We know how to contract rational curves of self-intersection -1 from Castelnuovo's Contractibility Criterion. How do we contract other curves that intersect K_S negatively?

It turns out that the answers to both of these questions will unite much of our exposition of minimal surfaces, and give us an algorithm to determine whether or not a surface is ruled/rational or not. Both answers stem from Mori's foundational paper, *Threefolds who's Canonical Bundles are not Numerically Effective* [Mor80], where the general theory of finding and contracting K_X-negative curves that has permeated much of the birational classification of higher-dimensional varieties first developed.

6.2 The Minimal Model Program for Surfaces: The Idea

Our goal is to contract the curves with negative intersection with the canonical bundle. In this light, the natural object of study is the space of 1-cycles, introduced in Section 2.1.5, modulo numerical equivalence (since the data we care about is the intersection number with K_X). To make things nicer, we will tensor this space with $\mathbb{R}$ to give it the structure of a vector space.

Definition 12. We define $N_1(X)$ to be the group of 1-cycles $Z_1(X)$ modulo numerical equivalence, tensored with $\mathbb{R}$ over $\mathbb{Z}$. Explicitly, these are formal sums of the form

$$\sum_{i=1}^{n} r_i [C_i]$$

where r_i are real numbers and C_i are numerical equivalence classes of curves.

Numerical equivalence of curves C_1 and C_2 means that for every divisor $D \in \mathrm{Div}(X)$ we have that $C_1.D = C_2.D$. The definition of numerical equivalence for 1-cycles extends directly from this. Also note that in the case of surfaces, the space of 1-cycles is just (dual to) the space of divisors, since codimension 1 varieties in a surface are just curves. However, we will use the language of 1-cycles so that the ideas generalize more nicely to higher dimensions. By the fact that numerical equivalence and homological equivalence agree up to torsion for curves in a smooth surface, we have that

$$N_1(X)^* \cong NS(S) \otimes_{\mathbb{Z}} \mathbb{R}$$

Note that in the general setting of complex projective varieties, this is also true with the appropriate generalizations of the Lefschetz 1-1 theorem and the arguments in Corollary 1. By the fact that $NS(S) \subset H^2(S, \mathbb{Z})$ and is thus finitely generated, we conclude that N_1 is a finite vector space of rank that of $NS(S)$, known as the *Picard number* $\rho(S)$ of S. Inside of this vector space, we are particularly interested in the space of effective 1-cycles, as these are the ones who's intersection with K_S we care about.

Definition 13. We define the *cone of curves* $NE(X)$ to be the subset of $N_1(X)$ who have representatives of the form

$$\sum_{i=1}^{k} r_i [C_i]$$

for C_i curves and r_i positive real numbers.

Warning: Note that it may be the case that D is an effective 1-cycle, but for a particular choice of basis $[C_1], \ldots, [C_n]$ for $N_1(X)$, writing $[D]$ as $\sum_{i=1}^{k} r_i [C_i]$ may require some r_i to be negative.

Effective classes are closed under scalar multiplication by positive real numbers and are closed under addition, so the set $NE(S)$ defines a convex cone in $N_1(S)$ (justifying the naming). It turns out to be much nicer for us to work with the closure of $NE(X)$, denoted $\overline{NE}(X)$, under some norm on $N_1(X)$. This is independent on the choice of norm as all norms on $\mathbb{R}^n$ are equivalent.

We are now in a position to answer our two questions from earlier. We are interested in a particular region of $\overline{NE}(X)$, the effective classes that lie to the negative side of the hyperplane $K_S^{\perp} = \{Z \in N_1(S) : Z.K_S = 0\}$. If we just take any 1-cycle Z lying on $\overline{NE}(X)$ in the half-space we're interested in, there's no guarantee that we can find a morphism from S that contracts the curve Z while not ruining S in the process. The *Contraction Theorem*, the first of two major theorems that we introduce, says that if C is a K_S-negative curve that is extremal in $\overline{NE}(S)$, then it can be contracted in a "non-disruptive" way.

More precisely, we define a half line ℓ as an *extremal ray* in a convex cone if for any two 1-cycles Z_1, Z_2 in $\overline{NE}(C)$, if $Z_1 + Z_2 \in \ell$, then both Z_1, Z_2 are in ℓ. See Figure 6.1 for an illustration of this concept for a cone in $\mathbb{R}^3$.

Then the Contraction Theorem is as follows:

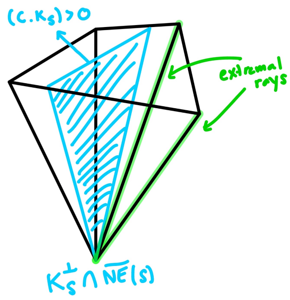

Figure 6.1: The extremal rays of a polygonal cone

Theorem 20. (The Contraction Theorem) Let X be a smooth projective variety and $\mathbb{R}_+[C]$ be a K_X-negative extremal ray generated by a curve C. Then there exists a normal projective variety Y and a morphism $\varphi : X \longrightarrow Y$ with connected fibers such that a curve is contracted to a point by φ if and only if the curve is in $\mathbb{R}_+[C]$. Both Y and the morphism are unique (up to isomorphism).

The fact that this theorem guarantees the contraction of precisely the effective curves in the numerical equivalence class (equivalently, homology class) of a curve C is extremely promising in our journey to contract K_S-negative curves without ruining our surface S in the process. This concept aligns with the contraction of rational curves of self-intersection -1 in Castelnuovo's Contractibility Criterion, and in fact these -1 curves will always be extremal in $\overline{NE}(S)$ as we will show. However, there are other curves that can represent an extremal ray. For instance, we will see that the fiber F of a geometrically ruled surface $S = \mathbb{P}_C(E)$ generates an extremal ray in the cone of curves. The Contraction theorem then produces a map that contracts the homology class of F and preserves the section h of $\mathcal{O}_S(1)$, which gives exactly the projection map of S to it's base curve C. In this way, we see that contracting a curve generating K_X negative extremal ray is somehow capturing more than just contracting -1 curves, but is "reconstructing" the structure of geometrically ruled surfaces as well!

Now the extremal rays we are searching for are in the *closure* of $NE(S)$. We are hoping to find a curve that represents a K_S-negative extremal ray. But why should a K_S-negative extremal ray have a representative by an effective curve? Let's work in the setting of $N_1(S) \cong \mathbb{R}^2$ (such as in the case of ruled surfaces over $\mathbb{P}^1$). Now suppose that an edge of the cone of curves has irrational slope. Then there can't even be an integral class (an element of N_1 with all r_i integers) as a representative of this extremal ray, much less an effective curve. Even if the slope of this edge is rational, there's no guarantee whatsoever that any integral class representing it is actually effective (see the warning after Definition 13). This is where our second major theorem, the *Mori Cone Theorem*, comes in. Mori's Cone Theorem not only tells us that for K_S-negative extremal rays there is a representative given by an effective curve C, but this curve can be chosen to be extremely nice:

Theorem 21. (The Mori Cone Theorem) Let X be a smooth projective variety of dimension n. Then every K_X-negative extremal ray has a representative given by a rational curve C satisfying

$$-1 \geq K_X.C \geq -n - 1$$

With Theorems 20 and 21, we can now define the Minimal Model program for surfaces, which serves to find a "nice" representative in each birational equivalence class (where "nice" in the surface case will turn out to mean minimal in our original definition). It may help to follow the flowchart in Figure 6.2 while reading the next paragraph.

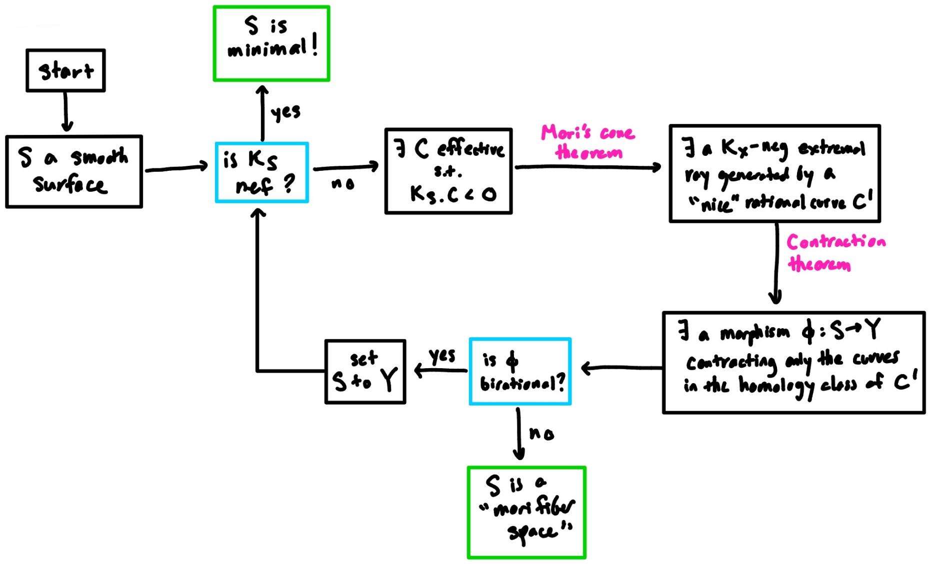

Figure 6.2: Running the Minimal Model Program for surfaces

We start with a smooth surface S. If K_S is nef to begin with, then we have found our "nice" representative (it's the unique minimal model in the birational equivalence class), and the program terminates. Otherwise, there is some curve C obstructing K_S from being nef, and by Mori's Cone Theorem, we can choose a rational curve C that generates an extremal ray. The Contraction Theorem gives us a morphism $\phi : S \to Y$ which contracts exactly the curves in the homology class of $[C]$ and no others. How we continue depends on what ϕ is:

- If ϕ is birational, then we can show that ϕ must be just blowing down a -1 curve, the contraction given in Castelnuovo's Contractibility Criterion. Then we update Y to be S and run the program plugging in Y.

- If ϕ is not birational, then we have represented S as a "Mori fiber space," a special kind of fibration that has lots of nice properties (that we will not get into here). In the case of surfaces, we will show that the only kinds of fiber spaces we can get are $\mathbb{P}^1$ bundles over a curve or $\mathbb{P}^2$ over a point. At this point, the program terminates.

In this way, we see that running the Minimal Model Program on a smooth surface is able to recover the structure of minimal models that we've derived over the past several chapters!

Even more miraculously, when constructing the minimal models of ruled and rational surfaces, the Minimal Model Program "knows" to contract all -1 curves before it contracts a fiber! Imagine we had a geometrically ruled surface S where we blew up a point on a fiber F_1. Now the worry we might have is that the Mori program looks to contract F_2, a fiber of S that we didn't blow up, before contracting the exceptional curve we just created by blowing up. Contracting F_2 (which is K_S-negative by adjunction) without contracting $\widehat{F_1}$ or E looks like it be tragically disruptive to our surface S. But when we blew up

F_1, F_2 is no longer extremal - it's of the form $\widehat{F_1} + E$ - so the program knows to wait to contract it! See the first two images in Figure 6.3 for an illustration.

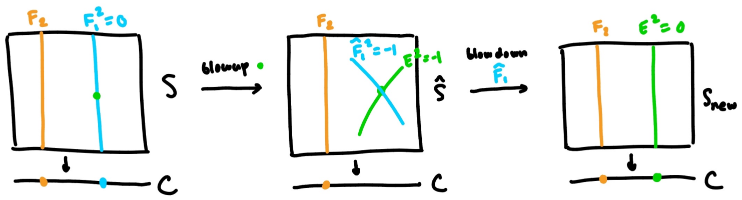

Figure 6.3: Blowing up and blowing down a fiber

Note that starting with $\widehat{S}$, we have the option to contract either E or $\widehat{F_1}$ first, and doing these in different orders gives us different geometrically ruled surfaces that are birational to $\widehat{S}$. So the Minimal Model Program also captures the non-uniqueness of minimal models (with our original definition) of ruled and rational surfaces! While we won't show it here, one can show that starting with any geometrically ruled surface over $\mathbb{P}^1$, you can get to any other such surface by blowing up a fiber, contracting the strict transform of the fiber, and repeating, shown in Figure 6.3 (this move is called an *elementary transform*).

6.3 The Minimal Model Program for Surfaces: The Details

Now that we have both the Contraction Theorem and the Mori Cone Theorem, we go back to fill in some details from the previous section. In particular, we give a classification of the possible kinds of extremal rays that can arise and the contractions that they induce.

Theorem 22. (Classification of Extremal Rays and Contractions for Surfaces) Let C be an irreducible curve on a smooth projective surface S such that $K_S.C < 0$. Then C is an extremal rational curve on $\overline{NE}(S)$ if and only if one of the following three statements hold:

1. $K_S.C = -1$, and C is a -1 sphere. In this case, the contraction $\varphi : S \longrightarrow S'$ is birational and S' is the blowdown of S (i.e. φ is the map constructed in Castelnuovo's Contractibility Criterion).

2. $K_S.C = -2$, S is the projectivization of a rank 2 vector bundle over a curve B, and C is a fiber of the standard projection map. In this case, the contraction $\varphi : S \longrightarrow B$ is the projection of S to the base curve.

3. $K_S.C = -3$, S is $\mathbb{P}^2$ and C is a line in $\mathbb{P}^2$. In this case, the contraction $\varphi : S \longrightarrow \mathrm{pt}$ is the constant map.

With this theorem along with Lemma 25, we recover the full results on the classification of minimal surfaces we derived in the previous chapters - that minimal models of irrational ruled surfaces are geometrically ruled over irrational curves, minimal models of rational surfaces are $\mathbb{P}^2$ and the Hirzebruch surface, and minimal models of all other surfaces are unique.

Corollary 8. Let S be a minimal surface (in the sense of no -1 curves), and K_S be not numerically effective. Then either X is $\mathbb{P}^2$ or X is a $\mathbb{P}^1$ bundle over a curve.

Proof. Since K_S is not numerically effective, there is an element of the cone of curves that lies in the half space $\{Z : K_S.Z < 0\}$. Then by Mori's Cone theorem, there is an K_S-extremal rational curve. The above theorem then immediately implies the result. □

This corollary also immediately shows that for S not ruled or rational, our new definition of minimality (K_S nef) agrees with our old one (no -1 curves).

We now go to give a proof of Theorem 22. We prove the backward direction first.

Classification of Extremal Rays, "if" part. We first show that in the three cases proposed in Theorem 22, that the rational curves given are in fact extremal rational in $\overline{NE}(S)$.

We start with rational curves of self-intersection -1. To show that an element C is extremal, it is sufficient to find a hyperplane in $N_1(S)$ that intersects $\overline{NE}(S)$ only at C. To find this, recall that in the proof of Castelnuovo's Contractibility Criterion we constructed a line bundle $\mathcal{L}$ on S that was very ample away from C and contracted C to a point. We can find a divisor of $\mathcal{L}$ that doesn't intersect C by considering the embedding $\varphi_{\mathcal{L}} : S \to \mathbb{P}^n$ with image the blowdown of S. Pulling back a hyperplane section not intersecting the point $\varphi_{\mathcal{L}}(C)$, we see that

$$\deg(\mathcal{L}|_C) = 0$$

For any other effective curve C' on $\widehat{S}$, since $\mathcal{L}$ is very ample we have that

$$\deg(\mathcal{L}|_{C'}) > 0$$

To see this, take a hyperplane section H intersecting C' transversely and not passing through $\varphi_{\mathcal{L}}(C)$. Then $\varphi_{\mathcal{L}}^* H.\varphi_{\mathcal{L}}^* C' > 0$, and so the degree of $\mathcal{L}$ on C is positive. Therefore, we get that the linear functional

$$\phi : \sum_{i=1}^{n} r_i [C_i] \mapsto \sum_{i=1}^{n} r_i \deg(\mathcal{L}|_{C_i})$$

is positive on all effective 1-cycles $NE(S)$ except on C where it's zero. Thus rational curves with self-intersection -1 generate extremal rays in $NE(S)$. Now we show that -1 curves generate extremal rays not just in $NE(S)$, but also $\overline{NE}(S)$. We will do this by showing that the zero hyperplane defined by the linear functional $\ell(\sum_i r_i C_i) := \sum_i r_i \deg(\mathcal{L}_{C_i})$ only intersects $\overline{NE}(S)$ in the extremal ray generated by C. For C_{ij} effective curves, suppose that a sequence of 1-cycles $Z_j = \sum_j r_{ij} C_{ij}$ converge in $N_1(S)$ and that they happen to converge to the zero hyperplane of ℓ. Then let Z'_j be Z_j minus any contribution of our -1 curve C. Clearly $\ell(Z_j) = \ell(Z'_j)$ for all j. Since ℓ evaluates to a positive integer for all $C_{ij} \neq C$, and the L^1 norm of any nonzero integral class of N_1 is bounded away from zero, we get that $\ell(Z'_j) \to 0$ implies that $||[Z'_j]|| \to 0$ by the triangle inequality. Putting these observations all together, we get that

$$Z_j = Z'_j + r_j C$$

for all j, with r_j nonnegative and Z'_j converging to zero. This means that Z_j must converge to rC for some nonnegative r, completing our original claim that -1 curves generate an extremal ray in $\overline{NE}(S)$. Finally, note by adjunction that we have that $K_S.C = -1$.

We now consider the case where S is a $\mathbb{P}^1$ bundle over a base curve B and F is a fiber of the standard projection map. We know from our study of the intersection theory of geometrically ruled surfaces that $N_1(S) \cong \mathbb{R}^2$ generated by $[F]$ and $[h]$ as defined in Theorem 15. We consider the ray $\mathbb{R}_+[F]$, which is in $\overline{NE}(S)$ since F is effective. This ray is extremal by checking by hand: if we have $a[F] + b[h]$ and $c[F] + d[h]$ that sum to be in $F^\perp \cap \overline{NE(S)}$, then we get $b + d = 0$. This implies either implies both $b, d = 0$ or one of b, d is negative. But b, d cannot be negative since otherwise we have that an effective divisor intersects F negatively, which is impossible. Therefore, both $b = d = 0$, and we get extremality. Finally, since $F^2 = 0$ adjunction gives us that $K_S.C = -2$.

Finally, we consider the case of $\mathbb{P}^2$ with the extremal ray being a line. In this setting, N_1 is just $\mathbb{R}$ and the ray generated by the line is positive half space, so is obviously extremal. Adjunction gives us that $K_S.C = -3$ since $C^2 = 1$. □

We now give a proof of the other direction, referring to arguments in the literature when needed. We will need a short lemma of Grothendieck:

Lemma 26. Every $D \in \overline{NE}(X)$ with positive self-intersection lies in the interior of $NE(X)$.

Proof. See Lemma 2.5 of Mori's paper [Mor80]. □

Classification of Extremal Rays, "only if" part. We show that if a curve C is an extremal rational curve on S, we must be in the three cases described in Theorem 22.

First suppose that $\rho(S) = \dim_{\mathbb{R}} N_1(S) \geq 2$. Now take the rational curve C representing our extremal ray. By the previous Lemma 26, $C^2 \leq 0$, since if $C^2 > 0$ it would have to be in the interior of $NE(S)$, which is nonempty for $\rho \geq 2$ and thus our curve could not be extremal, a contradiction. Since C is extremal, $K_S.C < 0$. By the genus formula for general curves (Corollary 3), we see that $p_a(C) \leq \frac{1}{2}$, and thus $p_a(C) = 0$. By the remark under Corollary 3, we conclude that C is smooth.

Now by adjunction we have that

$$-2 = K.C + C^2$$

There are two possibilities for the intersection behavior of C. One possibility is that we have

$$K.C = -1, C^2 = -1$$

This concludes that C is a -1 curve, and the uniqueness of the morphism in the Contraction Theorem forces the contraction to be the blowdown map of Castelnuovo's Contractibility Criterion. The other possibility is that

$$K.C = -2, C^2 = 0$$

In this case, we will show that for sufficiently large n the map $\varphi_{|nC|}$ corresponding to the line bundle $O_S(nC)$ is the contraction associated with the extremal ray $\mathbb{R}_+[C]$ (up to a finite morphism), and that the image of $\varphi_{|nC|}$ is dimension 1. First, note that $\mathcal{O}_S(nC)$ is basepoint-free[1]. An effective curve gets contracted to a point under $\varphi_{|nC|}$ if and only if it's intersection with C is zero. Since $h^0(S, \mathcal{O}_S(nC)) \geq 2$ and $C^2 = 0$, larger than a codimension 1 set in S gets contracted and therefore Y cannot be dimension 2. If Y is dimension zero, φ contracts all homology classes of Y. Then since $\mathrm{rk}NS(S) \geq 2$, φ contracts a class that is not $[C]$, a contradiction. Thus Y must be a curve. Since Y is normal by the contraction theorem, Y is smooth.

We now show that $[C]$ is the only numerical equivalence class contracted by $\varphi_{|nC|}$. Suppose that some numerical equivalence class $[D] \neq [C]$ for D an effective 1-cycle is contracted as well, thus $D.C = 0$ and D must be a fiber of $\varphi_{|nC|}$. D cannot be the class of an entire fiber, otherwise $[D] = [C]$, so the fiber containing D is of the form $D + D'$ for D' effective. Now $[D] + [D'] \in \mathbb{R}_+[C]$ but $[D]$ and $[D']$ are not in $\mathbb{R}_+[C]$, contradicting the extremality of $[C]$. By Stein factorization, we can find a φ that agrees with $\varphi_{|nC|}$, has connected fibers, and contracts the same curves, and so φ is the desired contraction.

Since Y is normal by the Contraction Theorem, Y is smooth, and now we have a map from smooth surface to a smooth curve with a $\mathbb{P}^1$ fiber. Then Noether-Enriques allows us to conclude that S is geometrically ruled over Y, C is a fiber of the projection map to Y, and the corresponding contraction to this extremal ray is just the projection map, completing the case where $\rho(S) \geq 2$.

In the case of $\rho(S) = 1$, the goal is to show that $h^1(S, \mathcal{O}_S) = 0$ and $h^0(S, \mathcal{O}_S(2K)) = 0$, from which we can run the exact argument of Theorem 16 to show that S must either be one of the Hirzebruch surfaces or $\mathbb{P}^2$. However, $\rho(S) = 1$ only matches with $\mathbb{P}^2$, and the proof would be complete. To show that $h^1(S, \mathcal{O}_S) = 0$, we use Kleiman's Criterion for Ampleness, which connects the ampleness of a line bundle to the cone of curves:

Lemma 27. (Kleiman's Criterion for Ampleness) A divisor D is ample if and only if for all elements in $\overline{NE}(S) - \{0\}$, $D.C > 0$.

We refer the reader to Proposition IV.7.5 in [Bar+04] for a proof of this lemma. By Kleiman's criterion we have that $-K_S$ is ample, and then $h^1(S, \mathcal{O}_S) = 0$ by the Kodaira vanishing theorem (see for instance 21.5.8 of [Vak25]). Since $-K_S$ is ample, so is $-2K_S$, and the Kodaira vanishing theorem again lets us conclude $h^2(S, \mathcal{O}_S(-K_S)) = h^0(S, \mathcal{O}_S(2K_S)) = 0$.

The contraction map associated to an extremal ray must contract the hyperplane class in $\mathbb{P}^2$, which just contracts all of $\mathbb{P}^2$ to a point. This completes the proof of Theorem 22. □

6.4 The Program for Threefolds and Beyond

The foundational issue in running the Minimal Model Program in dimension 3 and higher is that the contractions produced by the Contraction Theorem need not be smooth; it is only guaranteed that the

[1] To see this, consider the short exact sequence

$$0 \to \mathcal{O}_S((n-1)C) \to \mathcal{O}_S(nC) \to \mathcal{O}_C(nC) \to 0$$

By the fact that $H^1(C, \mathcal{O}_C(nC)) = 0$ and the fact that $H^1(S, \mathcal{O}_S(nC))$ stabilizes for large enough n, we get a surjection $H^0(C, \mathcal{O}_S(nC)) \to H^0(C, \mathcal{O}_S(nC))$ for large enough n. Since $\mathcal{O}_C(nC) = \mathcal{O}_C$ is basepoint-free, so is $\mathcal{O}_S(nC)$.

image of this map is normal. In the surface case, we get lucky - the contraction of -1 curves has smooth image by Castelnuovo's Contractibility Criterion, and this is the only birational morphism that can arise in the program. In dimension 3, the Contraction Theorem can produce morphisms from X to a variety Y of a lower dimension, which again gives us the representation of X as a Mori fiber space. However, in the case that the Contraction Theorem produces a birational morphism $\phi : X \to Y$ of the same dimension, the program asks us to plug Y back into the program. But Y *may now have singularities*! This introduces substantial complications to the program. The key insight of Mori and others in the development of the Minimal Model Program was that the Contraction Theorem and Mori Cone Theorem can be proven to hold not just for smooth varieties, but varieties with mild enough singularities. Then by allowing these kinds of singularities in the representatives we are searching for, we can still preserve the general shape of the program.

There are the two kinds birational contractions that can occur for projective varieties of dimension 3 and higher:

- If the exceptional set of $\phi : X \to Y$ is a divisor in X, i.e. codimension 1, then we call this a *divisorial contraction*. Divisorial contractions produce singularities that are mild enough to be fed back through our two key theorems, and so we can update X to Y and rerun the program.
- If the exceptional set of $\phi : X \to Y$ is codimension 2 or greater in X, then we call this a *small contraction*. These kinds of contractions can introduce extremely bad singularities that are too complicated for the Mori Cone Theorem and the Contraction Theorem to hold. To circumvent this and continue running the program, one needs to construct the *flip* of X, a variety X' birational to X constructed by performing a particular algebraic codimension 2 surgery. We plug X' back into the program, and hope that when we run the Contraction Theorem we get a morphism that doesn't induce a small contraction.

See Figure 6.4 for a flowchart of the program for threefolds and higher dimensional varieties.

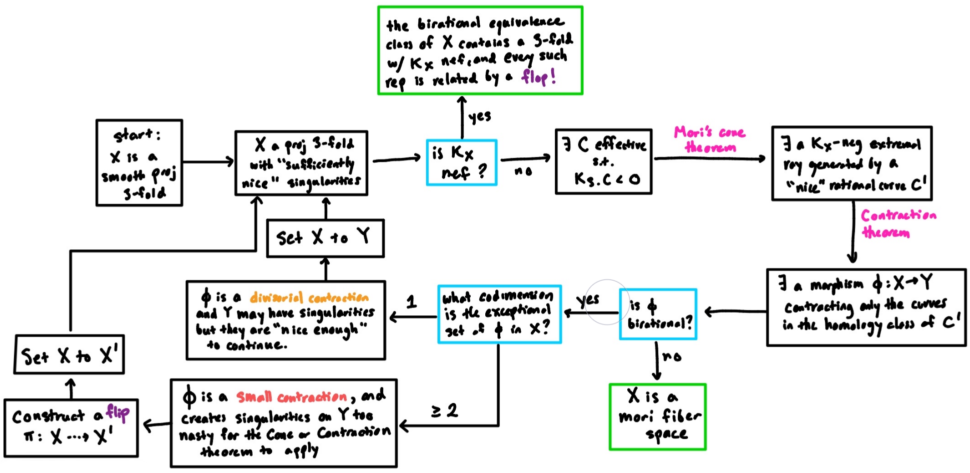

Figure 6.4: Running the Minimal Model Program for threefolds

It is not clear at all that the program terminates in a finite sequence of flips. In fact, this has only been proved for threefolds, and the termination of the Minimal Model Program in dimensions four and higher are still open.

While the definition of minimality given in Definition 11 behaves differently in dimensions 3 and higher, it still carries across some semblance of uniqueness of minimal models that we saw in the case of surfaces. In the complicatedness of high-dimensional varieties and their birational equivalence classes, it's unreasonable to expect that if a birational equivalence class has a representative with K_X nef, then it should be the only one up to isomorphism. However, we get something almost as nice: every such variety with K_X nef are related by a sequence of *flops*, a specific kind of algebraic codimension 2 surgery.

In both of these ways, we see that birational geometry beyond the surface case can become significantly more complicated. For a proper treatment of the Minimal Model Program in higher dimensions and the singularities that can be introduced in the program, see any of the surveys mentioned in the beginning of this chapter, as well as the foundational textbook of Kollár and Mori [KM98].

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